Comparative Analysis of FIR and IIR Filters for ECG Signal Denoising with DSP Implementation of the Selected Filters

A Report submitted in the partial fulfilment of the requirement for

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in

Electronics and Communication Engineering

Submitted by

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CERTIFICATE BY THE SUPERVISOR

This is to certify that this dissertation work titled **"Comparative Analysis of FIR and IIR Filters for ECG Signal Denoising with DSP Implementation of the Selected Filters"** being submitted by Purre H N S Chandra Shekar (23011A0414), Sibbina Bhavana (23011A0445). In partial fulfilment of the requirement for the award of the degree of Bachelor of Technology in **Electronics and Communication Engineering from JNTUH** during the academic year 2025-2026 is a record of bona fide work carried out by them, under my supervision. This dissertation has not been submitted to any other university or institution for the award of any other degree. The results are verified and found to be satisfactory.

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DECLARATION OF THE CANDIDATE

We, by Purre H N S Chandra Shekar (23011A0414), Sibbina Bhavana (23011A0445) hereby certify that The project report entitled **"Comparative Analysis of FIR and IIR Filters for ECG Signal Denoising with DSP Implementation of the Selected Filters"** is a bonafide record work done and submitted under the guidance of Dr. K. Anitha Sheela, Professor, Department of ECE, JNTUH, in particular fulfilment of the requirements for the award of the Degree of Bachelor of Technology in **Electronics and Communication Engineering** during the academic year 2025-2026. This is a record of bonafide work carried out by us and the results embodied in this project have not been reproduced/copied from any source and have not been submitted to any other University or Institute for the award of any other degree or diploma.

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We write this acknowledgement with great honour, pride, and pleasure to pay our respects to the people who enabled us either directly or indirectly in the accomplishment of this project.

We are immensely thankful to each and every faculty member of ECE Department for their relentless contribution towards the successful completion of our course.

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CONTENTS

ABSTRACT

Electrocardiogram (ECG) signals are essential for diagnosing cardiovascular diseases, but they are often corrupted by noise such as Baseline Wander (BW) and Power Line Interference (PLI), which can affect accurate analysis. This project focuses on the removal of these noise components using digital filtering techniques.

Clean ECG signals were obtained from the MIT-BIH Arrhythmia Database, while BW and PLI noise records were taken from the MIT-BIH Noise Stress Test Database. The selected noise signals were added to the clean ECG in MATLAB to generate noisy ECG signals. For each noise type, FIR and IIR filters were designed and evaluated in MATLAB. Their performance was analyzed using metrics such as Signal-to-Noise Ratio (SNR), Mean Squared Error (MSE), and error analysis to identify the most effective filter for BW and PLI removal.

The coefficients of the selected filters were exported as header files and implemented on the TMS320C6713 DSP Starter Kit using Code Composer Studio (CCS). The hardware results were compared with MATLAB simulations to validate the real-time implementation.

This project demonstrates an efficient approach for ECG signal denoising, which is crucial for accurate diagnosis and real-time biomedical signal processing applications.

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CHAPTER-1: INTRODUCTION

1.1 INTRODUCTION TO ECG SIGNAL

Electrocardiography (ECG) is a non-invasive diagnostic technique used to record the electrical activity of the heart. It is one of the most widely used methods for the diagnosis and monitoring of cardiovascular disease. ECG signal represents the electrical impulses generated during the contraction and relaxation of the heart muscles. These electrical activities are recorded using electrodes placed on the surface of the human body and are displayed in the form of a waveform.

ECG plays a vital role in the diagnosis and monitoring of various cardiovascular diseases such as arrhythmias, myocardial infarction, heart block, and ischemic heart diseases. Since the ECG signal contains valuable information regarding the functioning of the heart, accurate acquisition and analysis of ECG signals are essential in modern healthcare systems.

With the advancement of digital signal processing techniques, ECG signals can be processed efficiently for diagnosis, monitoring, and real-time healthcare applications. However, ECG signals are often contaminated by different types of noise during acquisition, transmission, and storage. Therefore, signal denoising has become an important step in ECG signal analysis.

1.2 ECG WAVE COMPONENTS

A standard ECG waveform consists of the P wave, QRS complex, T wave, and several intervals and segments, each representing a specific electrical event occurring during the cardiac cycle. These components provide important information regarding the electrical and physiological activities of the heart.

P Wave

The P wave represents the depolarization of the atria. It indicates the initiation of electrical activity that causes the atria to contract and pump blood into the ventricles.

QRS Complex

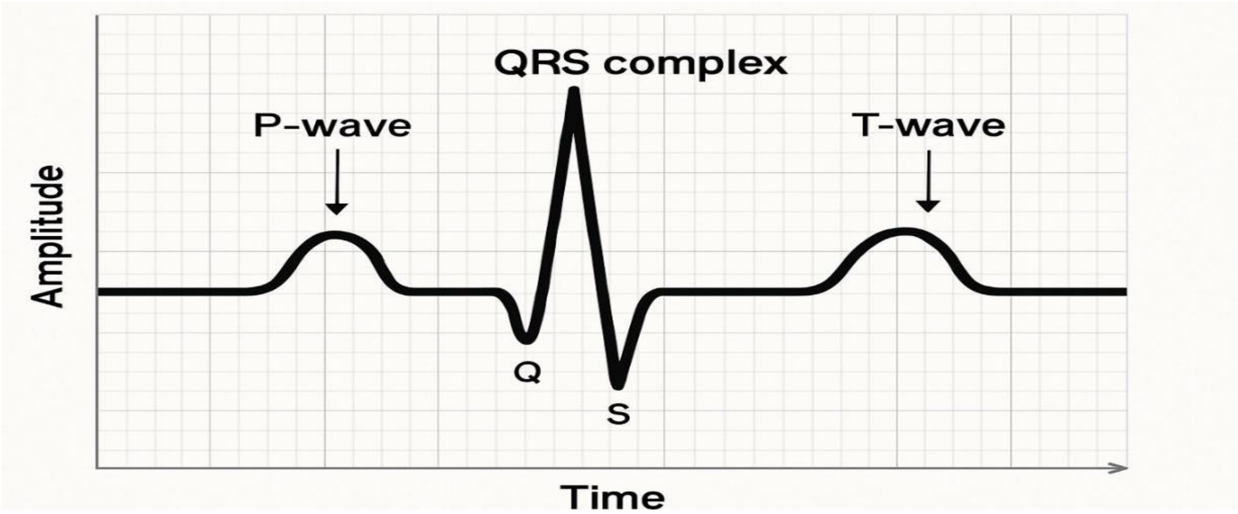
The QRS complex represents the depolarization of the ventricles. It is the most prominent portion of the ECG signal and corresponds to ventricular contraction. The QRS complex consists of three waves namely Q, R, and S waves.

T Wave

The T wave represents ventricular repolarization. It indicates the recovery phase of the ventricles after contraction and prepares the heart for the next cardiac cycle.

ECG Morphology

The shape, duration, and amplitude of these waves provide significant clinical information. Any distortion or alteration in these waveforms due to noise may lead to incorrect diagnosis and interpretation of cardiac conditions.



1.2 components of standard ECG signal

1.3 NEED FOR ECG SIGNAL DENOISING

The ECG signal is highly susceptible to various types of noise and interference generated during the recording process. These unwanted disturbances can significantly affect the quality of the signal and make clinical interpretation difficult.

Accurate diagnosis of cardiovascular diseases requires clean and noise-free ECG signals. Noise contamination can alter important ECG features such as wave amplitudes, intervals, and morphological characteristics. Consequently, automated diagnostic systems may produce inaccurate results.

ECG signal denoising aims to remove unwanted noise components while preserving the original morphological characteristics of the ECG waveform. Effective denoising improves signal quality, enhances diagnostic accuracy, and supports reliable real-time monitoring Systems .

Among the various denoising techniques, digital filters are widely employed because they provide effective noise suppression while preserving the important morphological characteristics of the ECG signal.

1.4 SOURCES OF NOISE IN ECG SIGNALS

During ECG acquisition, several types of noise can corrupt the recorded signal. These noises may originate from the patient, recording instruments, surrounding electrical equipment, or environmental conditions. The presence of noise reduces signal quality and may affect accurate diagnosis.

The major types of noise commonly present in ECG signals are:

1. Baseline Wander (BW)

A low-frequency noise caused by respiration, body movements, and electrode motion. It produces slow fluctuations in the ECG baseline.

2. Power Line Interference (PLI)

An electrical noise introduced from nearby power supply systems operating at 50 Hz or 60 Hz. It appears as a sinusoidal interference in the ECG signal.

3. Muscle Artifact Noise (EMG Noise)

This noise is generated due to the electrical activity of skeletal muscles. It is usually a high-frequency noise that overlaps with the ECG signal.

4. Electrode Motion Artifact

This noise occurs when electrodes move relative to the skin surface, causing abrupt variations in the recorded ECG signal.

5. Motion Artifact Noise

Patient movement during ECG recording can introduce unwanted disturbances and distort the signal waveform.

Among these noise sources, Baseline Wander (BW) and Power Line Interference (PLI) are considered the most common and significant disturbances affecting ECG recordings. These noises can obscure important ECG features and reduce the reliability of diagnosis. Therefore, this project focuses on the removal of Baseline Wander and Power Line Interference using various digital filtering techniques.

1.4.1 Selection of Baseline Wander and Power Line Interference

Although several types of noise affect ECG signals, this project focuses on Baseline Wander (BW) and Power Line Interference (PLI) which are common noise components using digital filtering techniques.

The reasons for selecting these noise components are:

• They frequently occur in practical ECG recordings.

• They significantly affect ECG signal quality and diagnostic accuracy.

• Their frequency ranges are well defined and predictable.

• Baseline Wander mainly exists below 0.5 Hz, whereas Power Line Interference occurs at a fixed frequency of 50 Hz or 60 Hz.

• Since the frequency components of BW and PLI are well defined and largely separable from the useful ECG spectrum, they can be effectively suppressed using appropriately designed digital filters.

• FIR and IIR filters can suppress these noises while preserving the important morphological features of the ECG signal.

• They are suitable for real-time implementation on DSP processors.

In contrast, other noise sources such as EMG noise, motion artifacts, and electrode motion artifacts have frequency components that overlap significantly with the useful ECG signal. Removing these noises using conventional filters may also remove important ECG information and distort the waveform. Therefore, advanced signal processing techniques are generally required for their suppression.

Hence, Baseline Wander and Power Line Interference were selected for detailed analysis and removal in this project.

1.4.2 Baseline Wander Noise

Baseline Wander is a low-frequency noise component that causes slow fluctuations in the baseline of the ECG signal. It appears as an upward or downward drifting of the ECG waveform.

The primary causes of Baseline Wander include:

- Respiration and breathing movements
- Body movements of the patient
- Electrode motion artifacts
- Poor electrode contact with the skin

The frequency range of Baseline Wander is typically below 0.5 Hz. Although it is a low-frequency disturbance, it can significantly affect the ST segment and other important ECG features. Therefore, removal of Baseline Wander is essential for accurate ECG analysis and interpretation.

1.4.3 Power Line Interference (PLI) Noise

Power Line Interference is caused by electromagnetic interference from nearby electrical power systems. In India, Power Line Interference typically occurs at a frequency of 50 Hz and becomes superimposed on the ECG signal.

Common sources of Power Line Interference include:

- Electrical equipment in hospitals and laboratories
- Power supply cables
- Monitoring instruments
- Electromagnetic radiation from surrounding devices

PLI appears as a sinusoidal noise component and can distort the ECG waveform, particularly affecting signal amplitude and frequency characteristics. Effective suppression of Power Line Interference is necessary to obtain reliable ECG recordings for clinical applications.

1.5 OBJECTIVES OF THE PROJECT

The primary objective of this project is to remove Baseline Wander Noise and Power Line Interference Noise from ECG signals using digital filtering techniques and implement the selected filters on a DSP processor.

The specific objectives are as follows:

* To acquire clean ECG signals from the MIT-BIH Arrhythmia Database.

* To obtain Baseline Wander (BW) and Power Line Interference (PLI) noise from the MIT-BIH Noise Stress Test Database and add them to the ECG signal.

* To design and implement various FIR and IIR filters for ECG denoising.

* To compare the performance of different filters using Signal-to-Noise Ratio (SNR), Mean Square Error (MSE), and Root Mean Square Error (RMSE).

* To identify the most suitable filter for Baseline Wander removal.

* To identify the most suitable filter for Power Line Interference removal.

* To implement the selected filters on the TMS320C6713 DSP processor.

* To compare the hardware implementation results with MATLAB simulation results to evaluate the performance of the selected ECG denoising filters.

1.6 SCOPE OF THE PROJECT

This project focuses on the denoising of Electrocardiogram (ECG) signals by removing two major noise components: Baseline Wander (BW) and Power Line Interference (PLI). Clean ECG signals are obtained from the MIT-BIH Arrhythmia Database, while the corresponding BW and PLI noise signals are obtained from the MIT-BIH Noise Stress Test Database. Various FIR and IIR digital filters are designed and evaluated for effective noise removal using MATLAB.

The performance of the designed filters is assessed using parameters such as Signal-to-Noise Ratio (SNR), Mean Square Error (MSE), and Root Mean Square Error (RMSE). Based on the comparative analysis, the most suitable filter for BW removal and the most suitable filter for PLI removal are selected. The selected filters are then implemented on the TMS320C6713 DSP processor using Code Composer Studio (CCS), and the hardware results are compared with MATLAB simulation results to validate the effectiveness of the proposed ECG denoising approach.

The scope of this project is limited to the removal of BW and PLI noise using digital filtering techniques. Other ECG noise sources, such as muscle artifacts (EMG), electrode motion artifacts, and motion artifacts, are beyond the scope of the present work.

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CHAPTER-2: LITERATURE REVIEW

2.1 EXISTING ECG DENOISING TECHNIQUES

ECG signals are affected by different types of noise, each having distinct frequency characteristics. These noise components can distort the ECG waveform and reduce the accuracy of clinical diagnosis. Over the years, several signal processing techniques have been proposed to remove these unwanted noise components while preserving the important characteristics of the ECG signal. The commonly used ECG denoising techniques are discussed below.

1. Wavelet Transform Based Methods

Wavelet Transform (WT) is one of the most widely used techniques for ECG signal denoising because of its excellent time-frequency localization capability. It decomposes the ECG signal into multiple frequency bands using scaling and wavelet functions. Noise components are removed by applying suitable thresholding techniques to the wavelet coefficients, after which the signal is reconstructed.

Wavelet-based methods are capable of effectively removing both high-frequency and low-frequency noise while preserving the important morphological features of the ECG waveform, such as the P wave, QRS complex, and T wave. These methods have been extensively used for biomedical signal processing applications due to their high denoising performance.

2. Adaptive Filtering Methods

Adaptive filtering is another widely used technique for ECG denoising, particularly in situations where the noise characteristics vary with time. Unlike fixed digital filters, adaptive filters continuously update their filter coefficients according to changes in the input signal and noise environment.

Among the various adaptive filtering algorithms, the Least Mean Square (LMS) and Recursive Least Square (RLS) algorithms are the most commonly used. The LMS algorithm updates the filter coefficients iteratively using the mean square error criterion, whereas the RLS algorithm provides faster convergence by minimizing the weighted least square error.

Adaptive filters are highly effective in removing correlated noise when an appropriate reference noise signal is available.

3. Artificial Intelligence Based Methods

Artificial Intelligence (AI) and Deep Learning (DL) techniques have recently gained significant attention for ECG signal denoising. These methods learn the complex relationship between noisy and clean ECG signals using large training datasets.

Deep learning models such as Convolutional Neural Networks (CNNs), Autoencoders, and Recurrent Neural Networks (RNNs) have demonstrated excellent denoising capability by automatically extracting relevant signal features without requiring manual feature engineering.

AI-based approaches can effectively suppress various types of noise while preserving the diagnostic information contained in the ECG signal.

4. Digital Filtering Methods

Digital filtering remains one of the most widely adopted techniques for ECG denoising because of its simplicity, computational efficiency, and ease of implementation. Digital filters are designed to suppress specific frequency components without significantly affecting the useful ECG information. Depending on their impulse response characteristics, digital filters are broadly classified into Finite Impulse Response (FIR) filters and Infinite Impulse Response (IIR) filters.

FIR Filter Based Approaches

Finite Impulse Response (FIR) filters produce an output that depends only on the present and previous input samples. Since they do not contain feedback, FIR filters are inherently stable and exhibit a linear phase response, making them highly suitable for biomedical signal processing applications where waveform preservation is important.

Several FIR filter design techniques have been proposed, including Window Method and Frequency Sampling Method. Window-based FIR filters such as Rectangular, Hamming, Hanning, Blackman, and Kaiser windows are widely employed for ECG denoising.

High-pass FIR filters are commonly used to suppress Baseline Wander, whereas notch FIR filters are employed to eliminate Power Line Interference around 50 Hz.

IIR Filter Based Approaches

Infinite Impulse Response (IIR) filters utilize both previous input samples and previous output samples through feedback connections. This feedback structure enables IIR filters to achieve sharp frequency responses using significantly lower filter orders than FIR filters.

Common IIR filter families include Butterworth, Chebyshev Type I, Chebyshev Type II, and Elliptic filters. High-pass IIR filters are generally employed for removing Baseline Wander, while notch IIR filters are widely used to suppress Power Line Interference.

Due to their lower computational complexity and reduced memory requirements, IIR filters are extensively used in real-time embedded and DSP applications.

2.2 SELECTION OF DIGITAL FILTERS FOR ECG DENOISING

Although several ECG denoising techniques have been proposed in the literature, each technique has certain limitations when considered for real-time implementation on Digital Signal Processing (DSP) hardware. These limitations motivated the selection of digital filtering techniques for the present work.

The limitations of the existing techniques are summarized below.

Wavelet Transform

- High computational complexity.
- Difficult to implement on real-time DSP hardware.
- Requires careful selection of wavelet basis functions and decomposition levels.
- Increased execution time for real-time applications.

Adaptive Filtering

- Requires a reference noise signal for effective operation.
- Convergence depends on algorithm parameters.
- Higher computational complexity compared to fixed digital filters.
- Performance may degrade under rapidly changing noise conditions.

Artificial Intelligence Based Methods

- Requires large labeled datasets for model training.
- High memory and computational requirements.
- Long training time.
- Difficult implementation on resource-constrained DSP processors.
- Reduced interpretability compared to conventional filtering methods.

In contrast, digital filters provide several advantages that make them highly suitable for the present work.

Advantages of Digital Filters

- Simple design and implementation.
- Low computational complexity.
- Easy implementation using MATLAB and Code Composer Studio.
- Suitable for real-time DSP applications.
- Effective removal of Baseline Wander and Power Line Interference.
- Easy comparison between different filter structures.

Easily implemented on real-time digital signal processing hardware.Based on the above discussion, digital filtering techniques offer a practical balance between denoising performance, computational efficiency, implementation simplicity, and hardware compatibility. Their ability to effectively suppress ECG noise while preserving important waveform characteristics makes them a suitable choice for real-time biomedical signal processing applications.

Table 2.3.1: Structural and Performance Comparison (FIR vs. IIR)

Parameter	**FIR Filter**	**IIR Filter**
Stability	Always Stable	May Become Unstable
Phase Response	Linear (No phase distortion)	Non-Linear (Phase distortion)
Filter Order	High (Requires more coefficients)	Low (More efficient cutoff)
Memory Requirement	High	Low
Computational Complexity	Higher	Lower

2.4 RESEARCH GAP

A comprehensive review of the existing literature indicates that considerable research has been carried out on ECG denoising using various signal processing techniques. However, several limitations still exist in current studies.

Most published research evaluates only a few filtering methods, making it difficult to perform a fair comparison of different digital filter structures. In addition, comparative studies involving both FIR and IIR filters under identical signal and noise conditions are relatively limited. Many investigations also focus primarily on software simulations, with comparatively less attention given to real-time DSP implementation and validation.

Consequently, there is a need for systematic comparative studies that evaluate multiple digital filtering techniques using common performance measures and assess their suitability for real-time biomedical signal processing applications.

Hence, there is a need for a comprehensive evaluation of multiple digital filtering techniques under identical conditions, followed by validation through real-time hardware implementation.

CHAPTER-3: SYSTEM DESIGN AND METHODOLOGY

3.1 SYSTEM OVERVIEW

The proposed ECG denoising system integrates software-based signal processing with hardware implementation to achieve effective removal of Baseline Wander (BW) and Power Line Interference (PLI) from ECG signals.

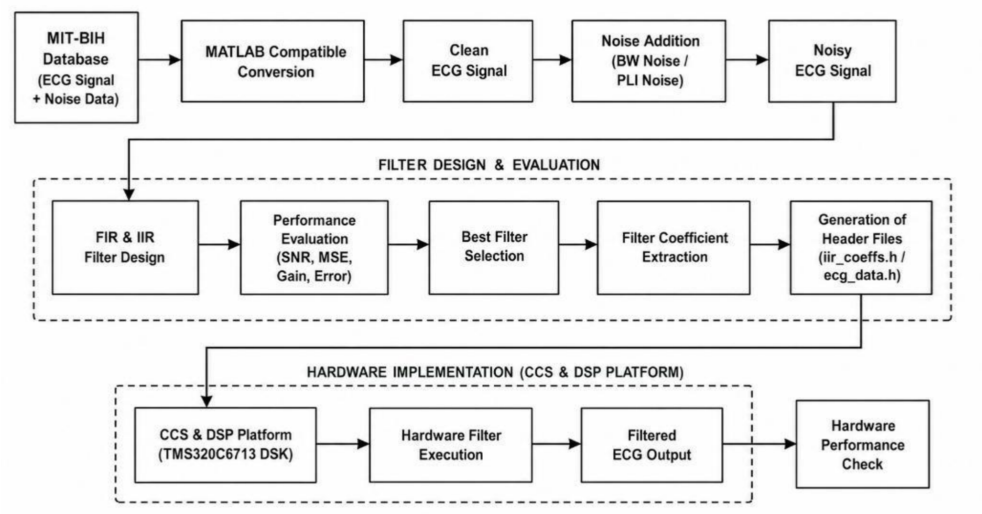


Figure 3.1- Overall System Architecture of the Proposed ECG Denoising System

Clean ECG signals are obtained from the **MIT-BIH Arrhythmia Database**, while the corresponding BW and PLI noise records are obtained from the **MIT-BIH Noise Stress Test Database**. The acquired data are converted into MATLAB-compatible format using Python preprocessing tools.

The selected BW and PLI noise signals are separately added to the clean ECG signal to generate noisy ECG signals for analysis. Various FIR and IIR digital filters are then designed and applied to the noisy ECG signals. The performance of each filter is evaluated using parameters such as Signal-to-Noise Ratio (SNR), Mean Square Error (MSE), Root Mean Square Error (RMSE), and gain to determine their denoising effectiveness.

Based on the performance evaluation, the most suitable filter for BW removal and the most suitable filter for PLI removal are selected. The coefficients of the selected filters are extracted and converted into CCS-compatible header files for hardware implementation. These header files, along with the ECG signal data, are used to implement the selected filters on the **TMS320C6713 DSK DSP processor** using **Code Composer Studio (CCS)**.

Finally, the filtered ECG output obtained from the DSP hardware is analyzed and compared with the MATLAB simulation results to validate the performance and effectiveness of the proposed ECG denoising system.

3.2 METHODOLOGY

The methodology adopted in this project consists of the following major phases:

Step 1: Data Acquisition

- Obtain the clean ECG signal from the MIT-BIH Arrhythmia Database.
- Obtain Baseline Wander (BW) and Power Line Interference (PLI) noise records from the MIT-BIH Noise Stress Test Database.

Step 2: Data Preprocessing

- Convert the acquired data into MATLAB-compatible format using Python and import them into MATLAB.

Step 3: Noise Addition

- Generate noisy ECG signals by separately adding BW and PLI noise to the clean ECG signal.

Step 4: Filter Design

- Design various FIR and IIR digital filters for ECG denoising.

Step 5: Performance Evaluation

- Evaluate the filters using SNR, MSE, RMSE, and gain.

Step 6: Best Filter Selection

- Select the most suitable filters for BW and PLI removal based on the evaluation results.

Step 7: DSP Implementation

- Generate CCS-compatible header files and implement the selected filters on the TMS320C6713 DSK DSP processor.

Step 8: Performance Validation

- Compare the hardware output with MATLAB simulation results to validate the effectiveness of the proposed ECG denoising system.

CHAPTER 4: SOFTWARE IMPLEMENTATION (in MATLAB)

4.1 ECG DATA ACQUISITION

The ECG signal used in this project was obtained from the MIT-BIH Arrhythmia Database. The selected record was converted into a MATLAB-compatible format and imported into MATLAB for processing.

This signal is used as the reference for noise addition, filter design, and performance evaluation.

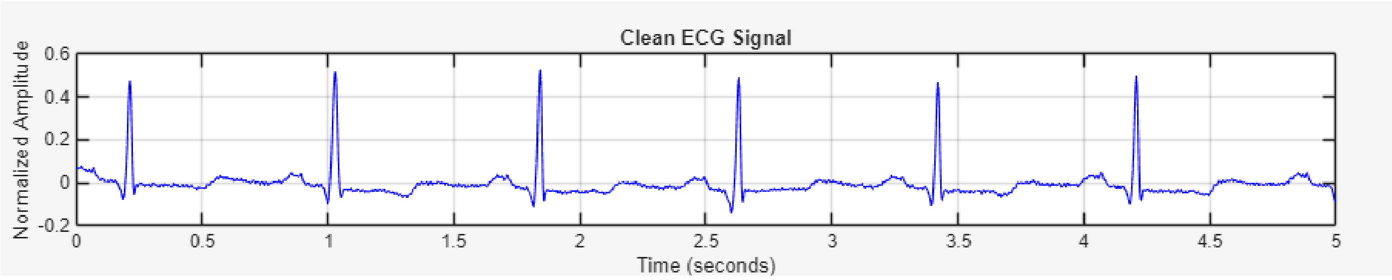


Figure 4.1: Original ECG Signal

4.2 GENERATION OF BASELINE WANDER NOISE

Baseline Wander (BW) is a low-frequency noise present in ECG recordings due to respiration, body movement, and electrode motion. Since the frequency components of BW noise are concentrated in the low-frequency region, it causes fluctuations in the ECG baseline and affects signal interpretation.

In this work, BW noise obtained from the MIT-BIH Noise Stress Test Database was added to the original ECG signal using MATLAB to simulate practical recording conditions.

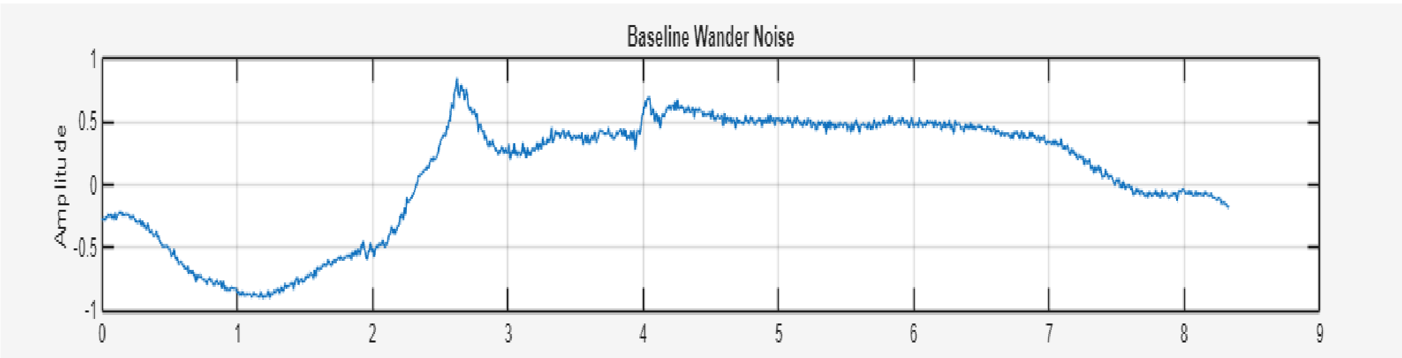


Figure 4.2.1 Baseline Wander Noise Signal

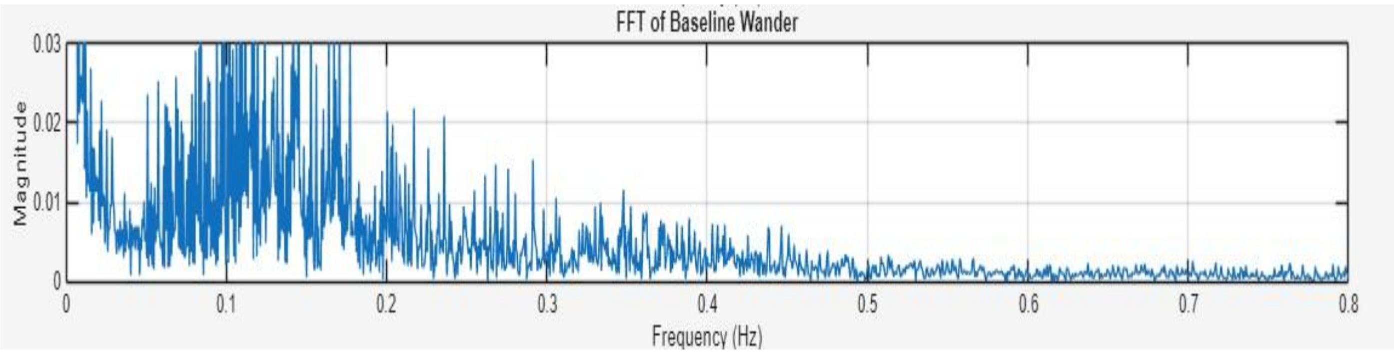


Figure 4.2.2 FFT Spectrum of Baseline Wander Noise

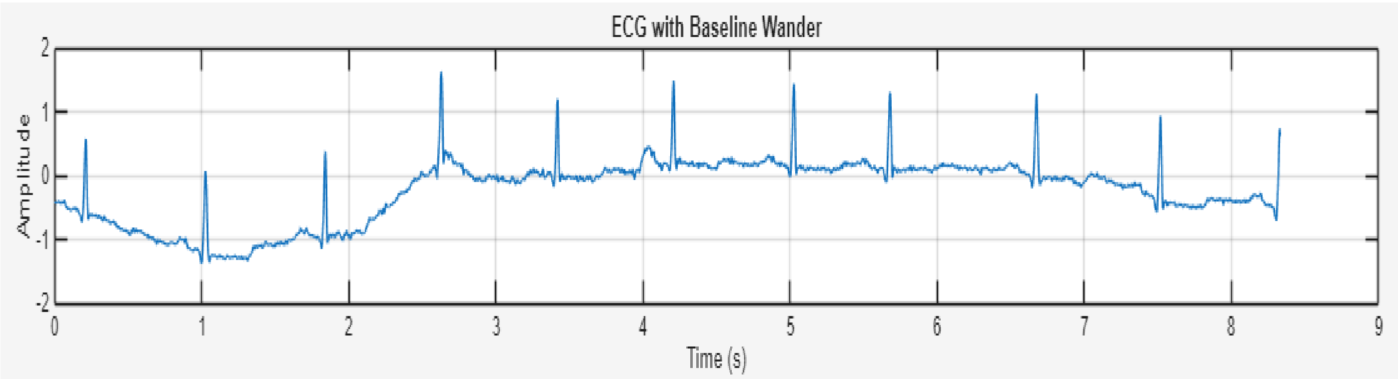


Figure 4.2.3 ECG Signal with Baseline Wander Noise

The FFT spectrum shows that the dominant frequency components of BW noise are concentrated below 0.5 Hz. Therefore, a high-pass filter with a cutoff frequency of 0.5 Hz was selected for BW noise removal. This cutoff frequency was kept constant throughout the study, while the filter order was varied to evaluate its effect on denoising performance.

It was observed that increasing the filter order improved the attenuation of low-frequency noise and produced a smoother ECG waveform. However, excessively high filter orders resulted in distortion of the ECG signal. The optimum filter order was selected based on the obtained SNR, MSE, and RMSE values. The selected specifications were then used for the implementation of IIR filters such as Butterworth, Chebyshev Type-I, and Chebyshev Type-II, as well as FIR filters based on Hamming, Hanning, Rectangular, Blackman, and Kaiser window techniques.

4.3 GENERATION OF POWER LINE INTERFERENCE NOISE

Power Line Interference (PLI) is a common noise present in ECG recordings due to electromagnetic interference from electrical equipment and power supply lines. This interference is primarily concentrated around 50 Hz and degrades the quality of the ECG signal.

In this work, PLI noise obtained from the MIT-BIH Noise Stress Test Database was added to the original ECG signal using MATLAB to simulate practical operating conditions.

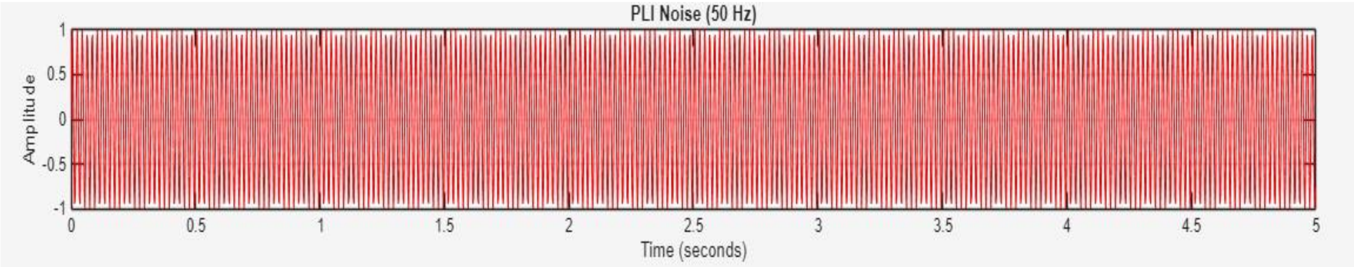


Figure 4.3.1 Power Line Interference Noise Signal

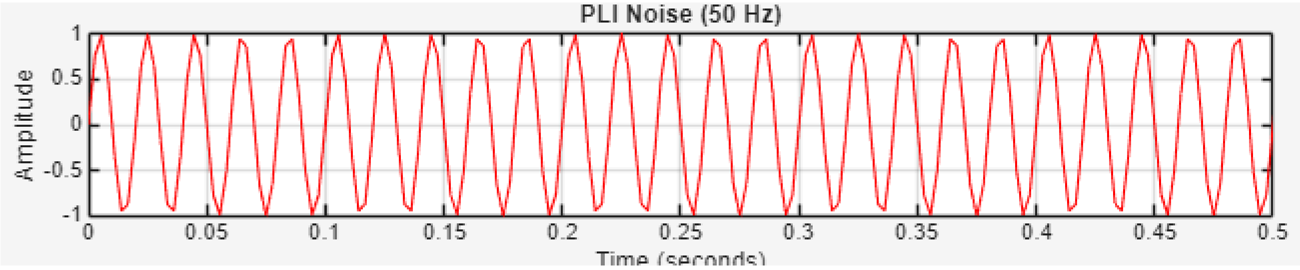


Figure 4.3.2 Power Line Interference Noise Signal (Zoomed Plot)

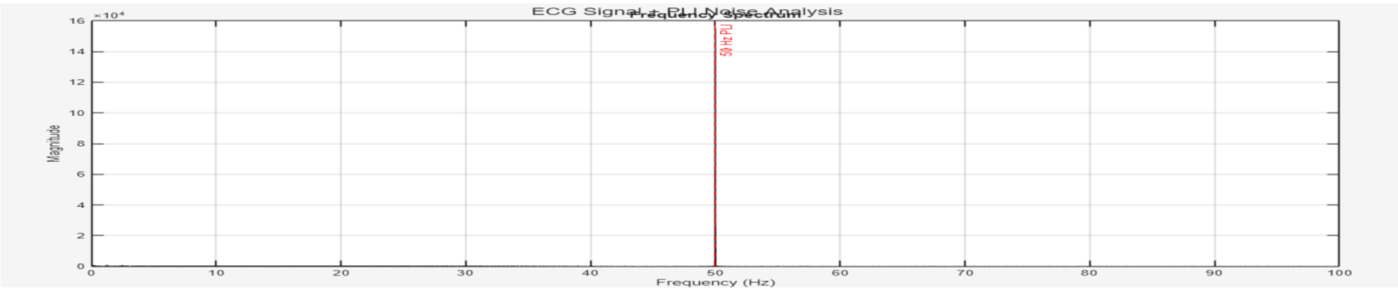


Figure 4.3.3 FFT Spectrum of Power Line Interference Noise

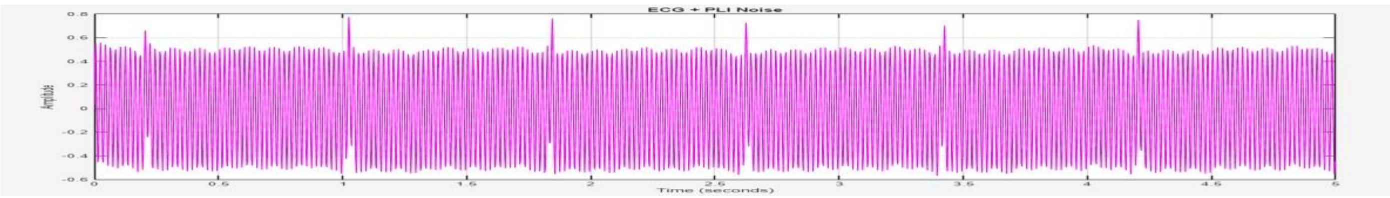


Figure 4.3.4 ECG Signal with Power Line Interference Noise

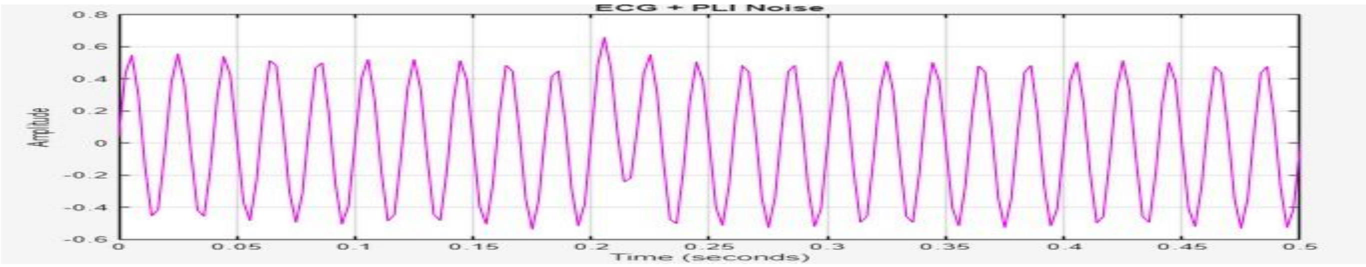


Figure 4.3.5 ECG Signal with Power Line Interference Noise(Zoomed Plot)

The FFT spectrum shows a prominent frequency component around 50 Hz, confirming the presence of power line interference. To suppress this noise, a bandstop filter centered at 50 Hz was designed. The lower and upper cutoff frequencies were selected as 45 Hz and 65 Hz respectively, resulting in a bandwidth of 20 Hz. These specifications were kept constant throughout the study, while the filter order was varied to evaluate the filtering performance.

It was observed that increasing the filter order improved the attenuation of the interference component and produced a response closer to the ideal bandstop filter. The optimum filter order was selected based on the obtained SNR, MSE, and RMSE values. The selected specifications were then used for the implementation of IIR filters such as Butterworth, Chebyshev Type-I, and Chebyshev Type-II, as well as FIR filters based on Hamming, Hanning, Rectangular and Blackman window techniques.

4.4 DESIGN AND IMPLEMENTATION OF IIR FILTER FOR BASELINE WANDER NOISE

IIR filters is used for the removal of **Baseline Wander (BW) noise** from the ECG signal.

IIR filter contains both poles and zeros, enabling sharp frequency responses with relatively lower filter orders. Due to their computational efficiency, IIR filters are widely used in real-time signal processing applications.

General IIR Filter Equation

$$\mathbf{y(n) = \sum_{k=0}^{M} b_k x(n-k) - \sum_{k=1}^{N} a_k y(n-k)}$$

where,

x(n) = Input ECG signal

y(n) = Filtered ECG signal

a_k and b_k = Filter coefficients

4.4.1 Butterworth High Pass Filter

The Butterworth filter was implemented in MATLAB for Baseline Wander noise removal. The filter was designed using the specifications discussed in Section 4.2 and applied to the noisy ECG signal

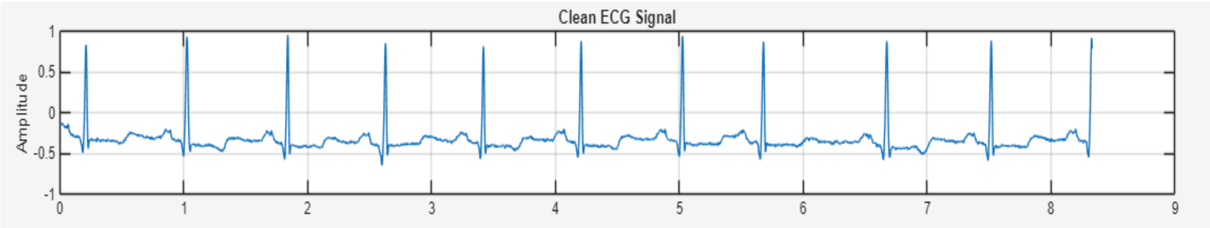


Figure 4.4.1(a) : Original ecg signal

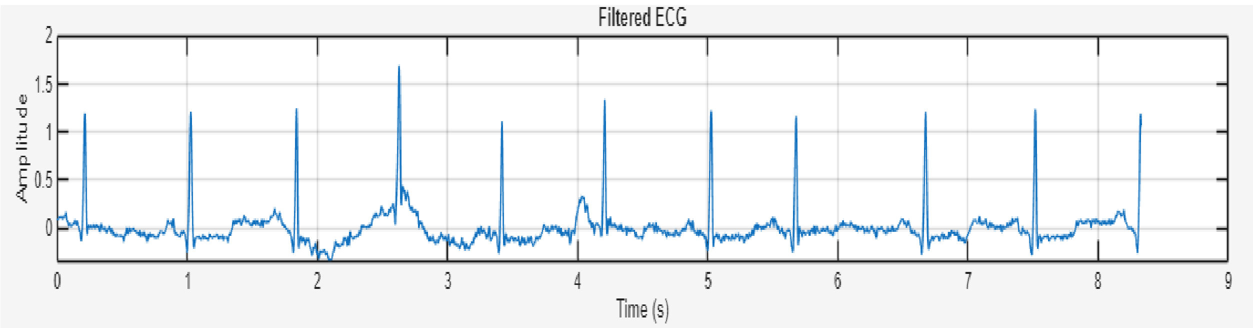


Figure 4.4.1(b): ECG Signal Filtered Using Butterworth Filter , order=8

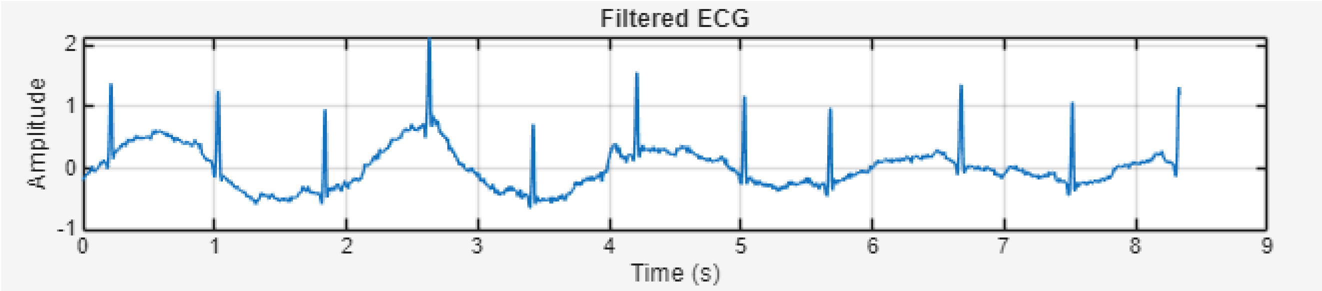


Figure 4.4.1(c): ECG Signal Filtered Using Butterworth Filter , order=20

Parameter	Value(order=8)	Value(order=20)
SNR	1.1910	1.1084
MSE	0.099689	0.1016
RMSE	0.31574	0.31876
Overall Gain	0.9779	0.9459

Table 4.4.1: Performance Parameters of Butterworth Filter

4.4.2 Chebyshev Type-I High Pass Filter

The Chebyshev Type-I filter was implemented in MATLAB for Baseline Wander noise removal. The filter was applied to the noisy ECG signal, and its performance was evaluated using SNR, MSE, and RMSE parameters.

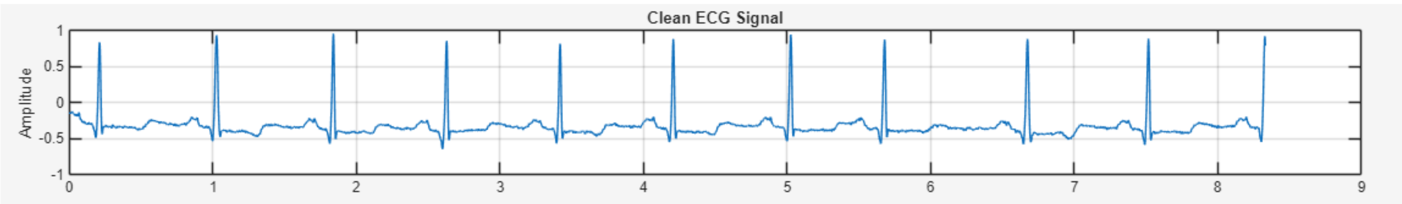


Figure 4.4.2(a): original ecg signal

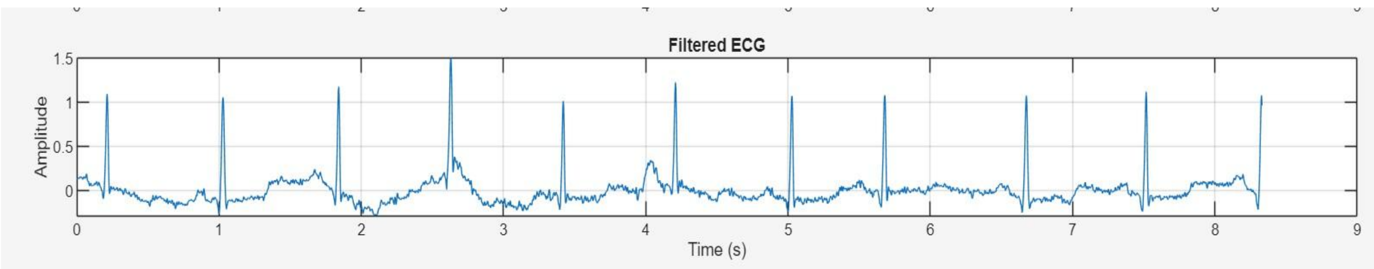


Figure 4.4.2(b): ECG Signal Filtered Using Chebyshev Type-I Filter, order -6

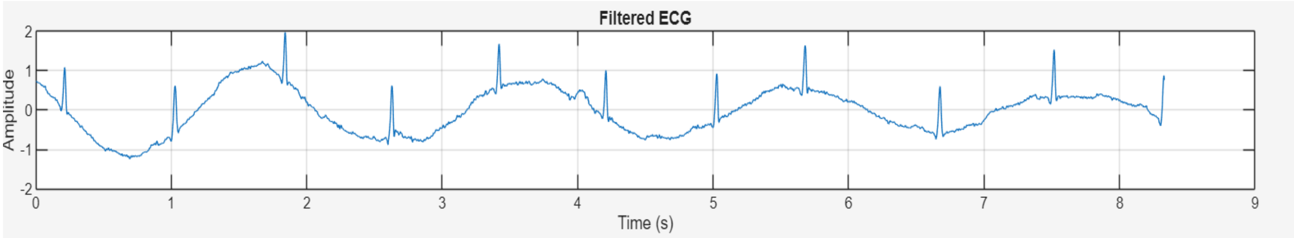


Figure 4.4.2(c) ECG Signal Filtered Using Chebyshev Type-I Filter, order -8

Parameter	Value(order-6)	Value(order-8)
SNR	1.1731	1.0060
MSE	0.1001	0.10403
RMSE	0.3163	0.32254
Overall Gain	0.9254	0.9178

Table 4.4.2 Performance Parameters of Chebyshev Type-I Filter

4.4.3 Chebyshev Type-II High Pass Filter

The Chebyshev Type-II filter was implemented in MATLAB for Baseline Wander noise removal. The filter was applied to the noisy ECG signal, and its performance was evaluated using SNR, MSE, and RMSE parameters.

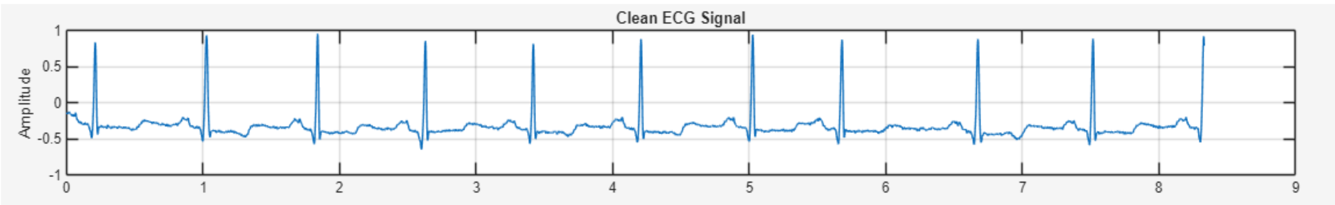


Figure 4.4.3(a): original ecg signal

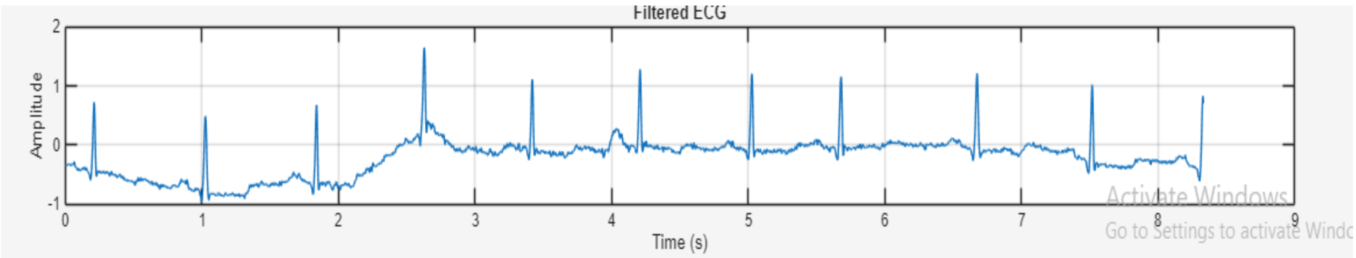


Figure 4.4.3(b): ECG Signal Filtered Using Chebyshev Type-II Filter , order-20

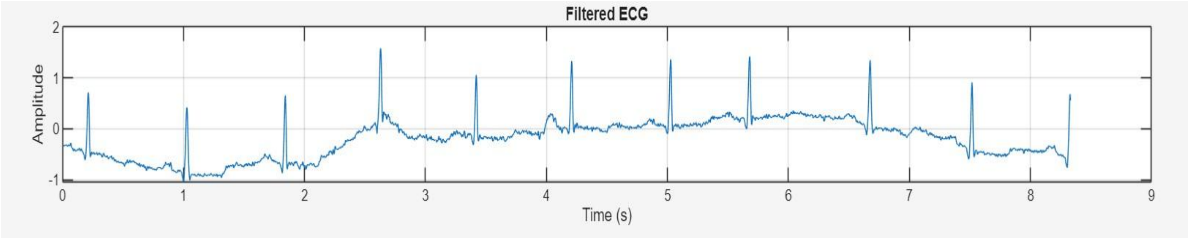


Figure 4.4.3(c): ECG Signal Filtered Using Chebyshev Type-II Filter , order-4

Parameter	**Value(order-4)**	**Value (order-20)**
SNR	-9.7452	-9.7345
MSE	1.2367	1.2337
RMSE	1.1121	1.1107
Overall Gain	0.9990	0.9990

Table 4.4.3: Performance Parameters of Chebyshev Type-II Filter

4.5 DESIGN AND IMPLEMENTATION OF FIR FILTERS FOR BASELINE WANDER NOISE

Finite Impulse Response (FIR) filters were used to remove Baseline Wander (BW) and Power Line Interference (PLI) from the ECG signal. They are inherently stable, as they contain only zeros, and provide a linear phase response that preserves ECG waveform characteristics.

FIR filters also allow straightforward coefficient implementation, making them suitable for real-time processing on the TMS320C6713 DSK processor.

Unlike IIR filters, FIR filters contain only zeros and no poles, which ensures unconditional stability during implementation.

General FIR Filter Equation

$$y(n) = \sum_{k=0}^{N} b_k x(n-k)$$

x(n) = Input ECG signal
y(n) = Filtered ECG signal
b_k = Filter coefficients

Window Functions Used

The FIR filters used in this work were designed using different windowing techniques, namely Hamming, Hanning, Rectangular, Blackman, and Kaiser windows. The corresponding window equations are given below.

Hamming Window

$$w(n) = 0.54 - 0.46\cos\left(\frac{2\pi n}{N-1}\right)$$

Hanning Window

$$w(n) = 0.5 - 0.5\cos\left(\frac{2\pi n}{N-1}\right)$$

Rectangular Window

$$w(n) = 1, 0 \leq n \leq N-1$$

Blackman Window

$$w(n) = 0.42 - 0.5\cos\left(\frac{2\pi n}{N-1}\right) + 0.08\cos\left(\frac{4\pi n}{N-1}\right)$$

4.5.1 Fir High Filter Using Hamming Window

The FIR filter using Hamming window was implemented in MATLAB for Baseline Wander noise removal. The filter was applied to the noisy ECG signal, and its performance was evaluated using SNR, MSE, and RMSE parameters.

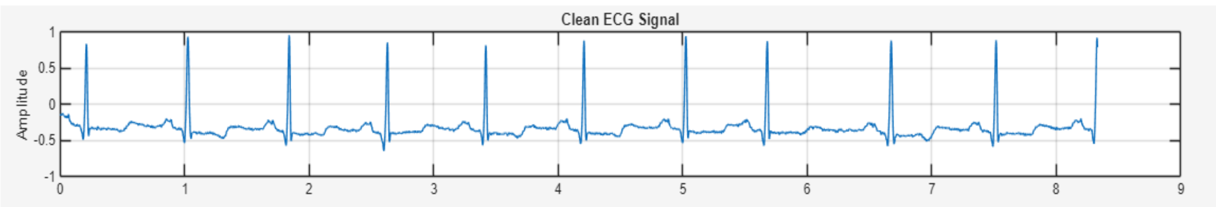


Fig: Original ECG Signal

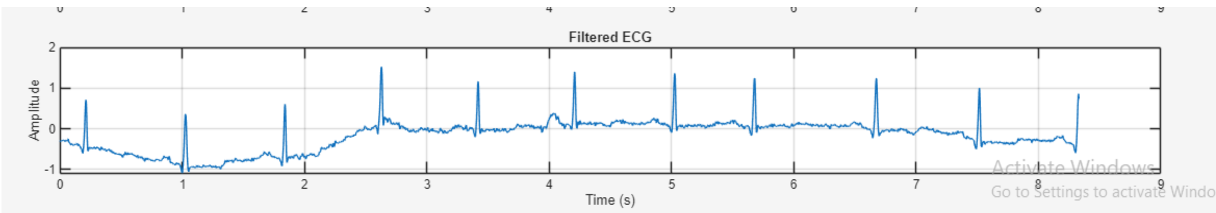


Figure 4.5.1(a): ECG Signal Filtered Using Fir Filter Using Hamming Window (order=100)

Parameter	Value
SNR	-8.5155
MSE	0.93175
RMSE	0.96527
Overall Gain	0.9990

Table 4.5.1 Performance Parameters of Butterworth Notch Filter

4.5.2 Fir High Filter Using Hanning Window

The FIR filter using Hanning window was implemented in MATLAB for Baseline Wander noise removal. The filter was applied to the noisy ECG signal, and its performance was evaluated using SNR, MSE, and RMSE parameters.

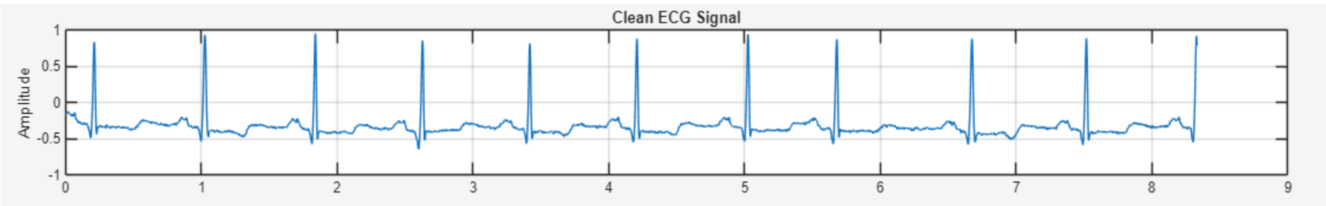


Fig: Original ECG Signal

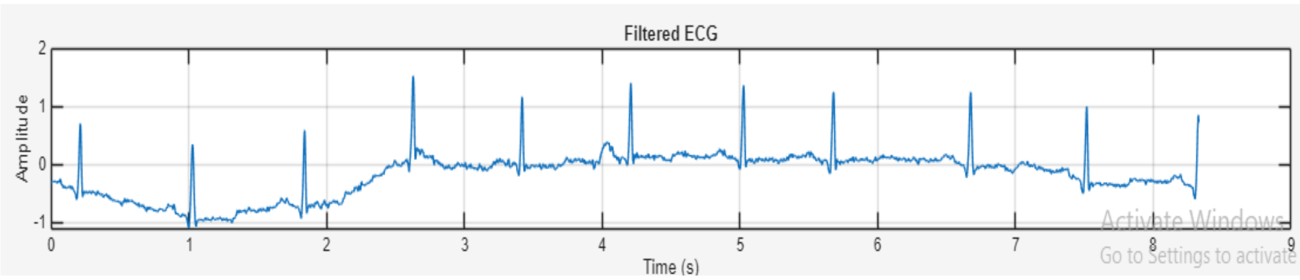


Figure 4.5.2(a): ECG Signal Filtered Using Fir Filter-Hanning Window (order=100)

Parameter	**Value**
SNR	-8.6439
MSE	0.95971
RMSE	0.97965
Overall Gain	2.232761

Table 4.5.2 :Performance Parameters of Chebyshev Type-I

4.5.3 Fir High Filter Using Rectangular Window

The FIR filter using rectangular window was implemented in MATLAB for Baseline Wander noise removal. The filter was applied to the noisy ECG signal, and its performance was evaluated using SNR, MSE, and RMSE parameters.

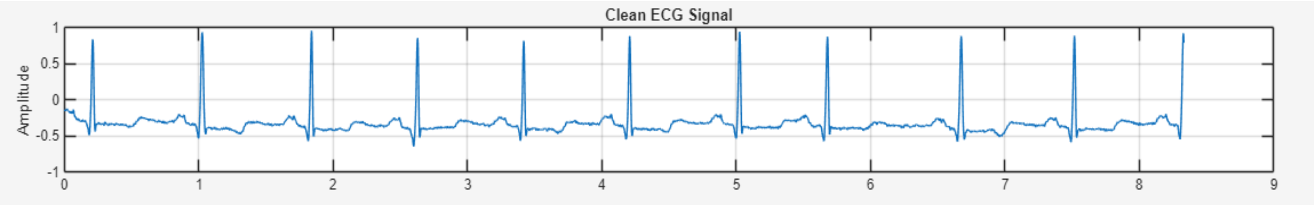


Fig: Original ECG Signal

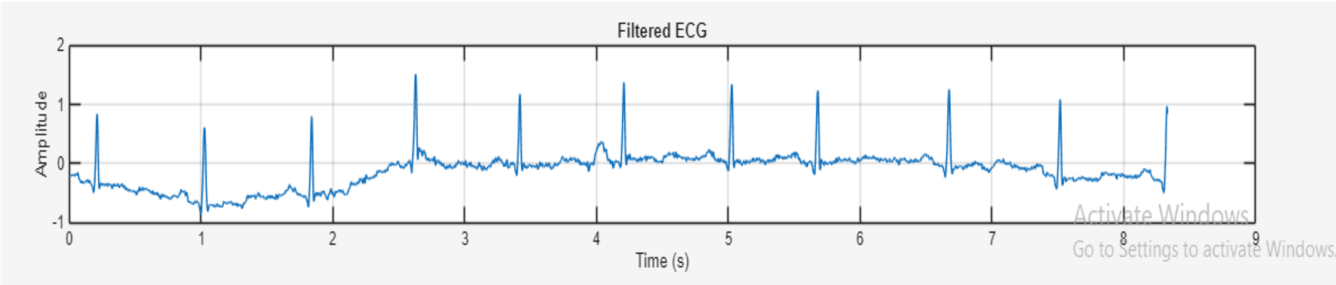


Figure 4.5.3(a): ECG Signal Filtered Using Fir Filter Using Rectangular Window(order=100)

Parameter	Value
SNR	-6.4851
MSE	0.58379
RMSE	0.76406
Overall Gain	1.630443

Table 4.5.3: Performance Parameters of Fir Filter Using Rectangular Window

4.5.4 Fir High Filter Using BLACKMANN Window

The FIR filter using rectangular window was implemented in MATLAB for Baseline Wander noise removal. The filter was applied to the noisy ECG signal, and its performance was evaluated using SNR, MSE, and RMSE parameters.

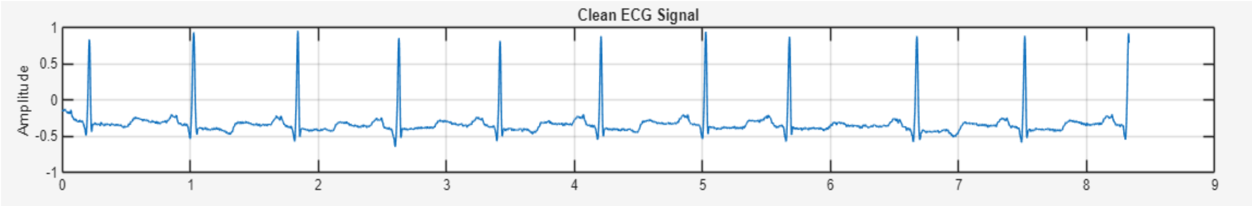


Fig: Original ECG Signal

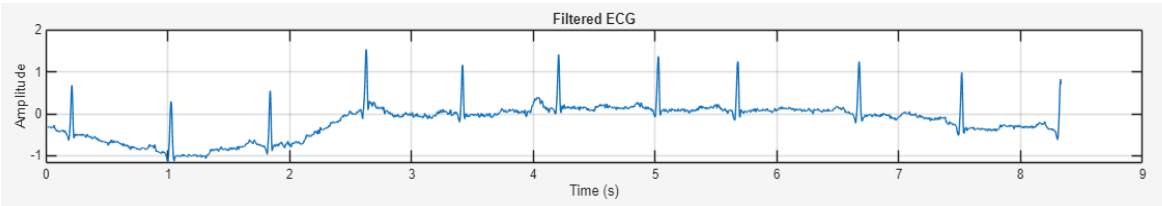


Figure 4.5.4(a): ECG Signal Filtered Using Fir Filter Using Blackmann Window (order =100)

Parameter	Value
SNR	-9.028
MSE	1.0485
RMSE	1.042
Overall Gain	2.356601

Table 4.5.4 Performance Parameters of Fir Filter Using Blackmann Window

OBSERVATIONS FOR BASELINE WANDER NOISE:

1. After comparing the performance of all the filters, the Butterworth high-pass filter was selected as the best filter for removing baseline wander noise from the ECG signal.
2. The Butterworth filter achieved the highest SNR of 1.1084 dB, indicating the greatest improvement in signal quality among all the filters considered.
3. The Chebyshev Type-I filter also showed good performance with an SNR of 1.0060 dB, but it was slightly lower than that of the Butterworth filter.
4. The FIR window-based filters (Hamming, Hanning, Blackman, and Rectangular) produced lower SNR values, indicating comparatively less effective baseline wander removal.
5. The Butterworth filter has a maximally flat passband response, which effectively removes low-frequency baseline drift while preserving the important ECG components.
6. Increasing the filter order improved the attenuation of baseline wander. However, beyond the optimum order, the improvement became insignificant while increasing computational complexity.
7. For ECG processing, it is essential to remove baseline wander without distorting the P wave, QRS complex, and T wave. The Butterworth filter achieved a good balance between noise suppression and waveform preservation.
8. Based on the SNR results and waveform quality, the Butterworth high-pass filter was selected as the most suitable filter for baseline wander removal and was implemented on the DSP hardwaress

POWER LINE INTERFERENCE NOISE REMOVAL

4.6 Design And Implementation Of IIR Filter For Power Line Interference Noise

General IIR Filter Equation

$$y(n) = \sum_{k=0}^{M} b_k x(n-k) - \sum_{k=1}^{N} a_k y(n-k)$$

where,
$x(n)$ = Input ECG signal
$y(n)$ = Filtered ECG signal
a_k and b_k = Filter coefficients

4.6.1 Butterworth Bandstop Filter

The Butterworth filter was implemented in MATLAB for Power line Interference noise removal. The filter was designed using the specifications discussed in Section 4.2 and applied to the noisy ECG signal.

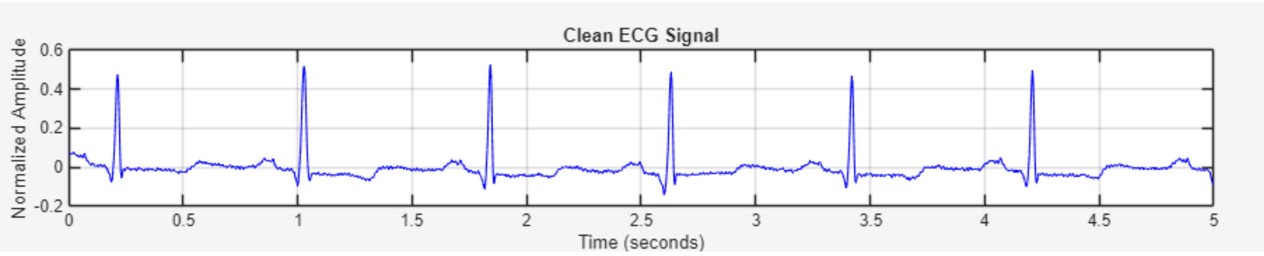


Fig: Original ECG Signal

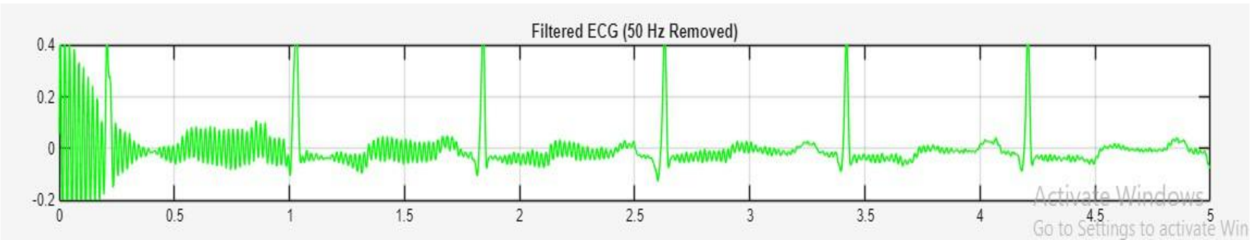


Figure 4.6.1(a): ECG Signal Filtered Using Butterworth Bandstop Filter(order =4)

Parameter	Value
SNR	21.2794
MSE	4.7918e-05
RMSE	0.0069
Overall Gain	0.930691

Table 4.6.1: Performance Parameters of Fir Filter Using Window

4.6.2 Chebyshev Type-1 Bandstop Filter

The Chebyshev Type-1 filter was implemented in MATLAB for Power line Interference noise removal. The filter was designed using the specifications discussed in Section 4.2 and applied to the noisy ECG signal.

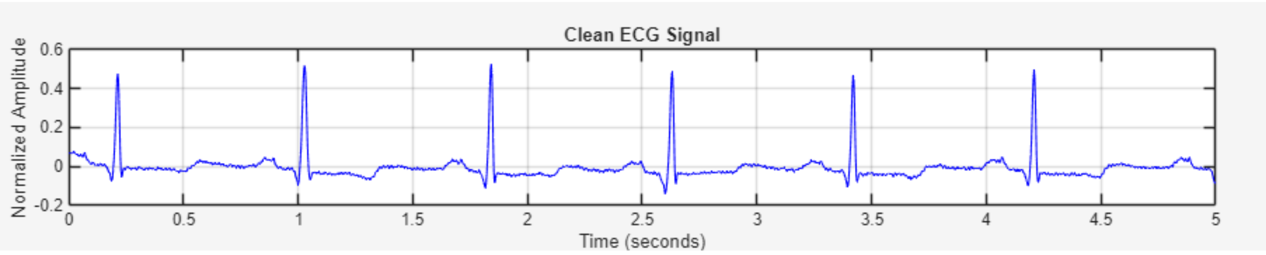


Fig: Original ECG Signal

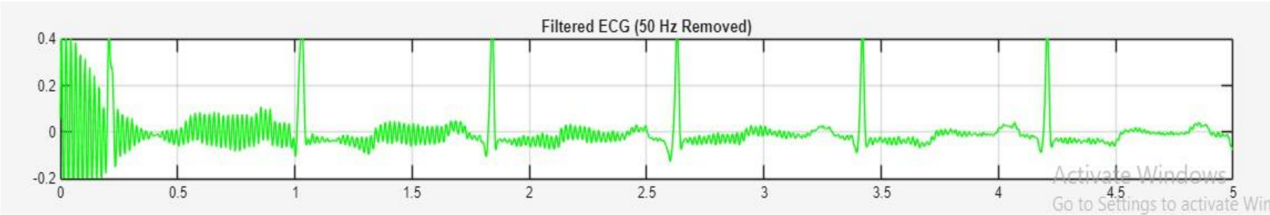


Figure 4.6.2(a): ECG Signal Filtered Using Chebyshev Type-1 Bandstop Filter (order=4)

Parameter	Value
SNR	19.2791
MSE	0.00007595
RMSE	0.00871504
Overall Gain	0.908094

Table 4.6.2: Performance Parameters of Chebyshev Type-1 Bandstop Filter

4.6.3 Chebyshev Type-2 Bandstop Filter

The Chebyshev Type-2 filter was implemented in MATLAB for Power line Interference noise removal. The filter was designed using the specifications discussed in Section 4.2 and applied to the noisy ECG signal.

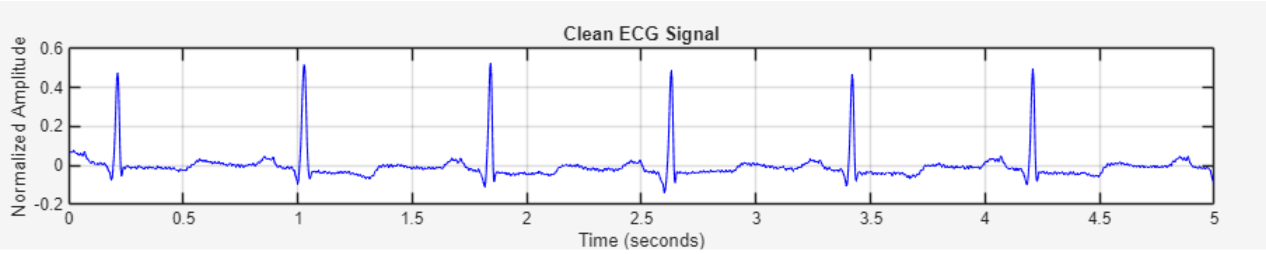


Fig: Original ECG Signal

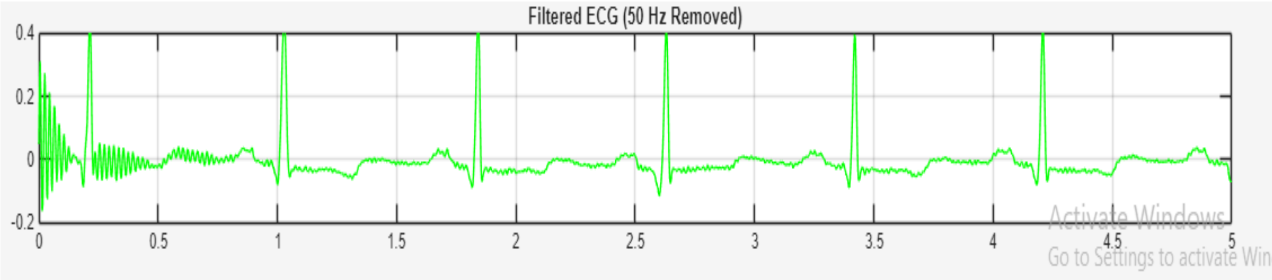


Figure 4.6.3(a): ECG Signal Filtered Using Chebyshev Type-2 Bandstop Filter (order=4)

Parameter	Value
SNR	16.4877
MSE	0.00014444
RMSE	0.01201814
Overall Gain	0.857472

Table 4.6.3: Performance Parameters of Chebyshev Type-2 Bandstop Filter

4.7 DESIGN AND IMPLEMENTATION OF FIR FILTERS FOR POWER LINE INTERFERENCE NOISE

Window Functions Used

The FIR filters used in this work were designed using different windowing techniques, namely Hamming, Hanning, Rectangular, Blackman. The corresponding window equations are given below.

Hamming Window

$$w(n) = 0.54 - 0.46\cos\left(\frac{2\pi n}{N-1}\right)$$

Hanning Window

$$w(n) = 0.5 - 0.5\cos\left(\frac{2\pi n}{N-1}\right)$$

Rectangular Window

$$w(n) = 1, 0 \leq n \leq N-1$$

Blackman Window

$$w(n) = 0.42 - 0.5\cos\left(\frac{2\pi n}{N-1}\right) + 0.08\cos\left(\frac{4\pi n}{N-1}\right)$$

4.7.1 Fir Bandstop Filter Using Hanning Window

The FIR filter using Hanning windowing was implemented in MATLAB for Power line Interference noise removal. The filter was designed using the specifications discussed in Section 4.2 and applied to the noisy ECG signal.

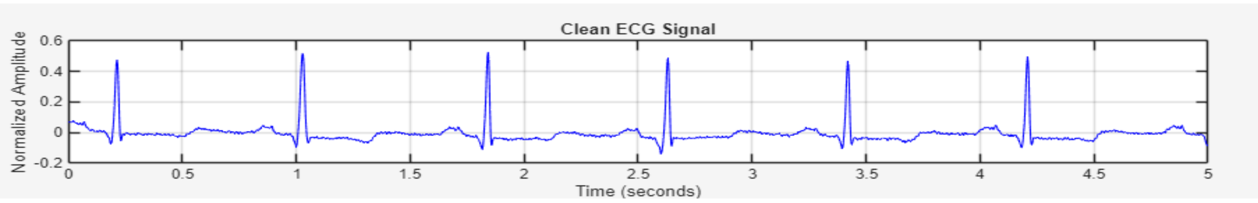


Fig: Original ECG Signal

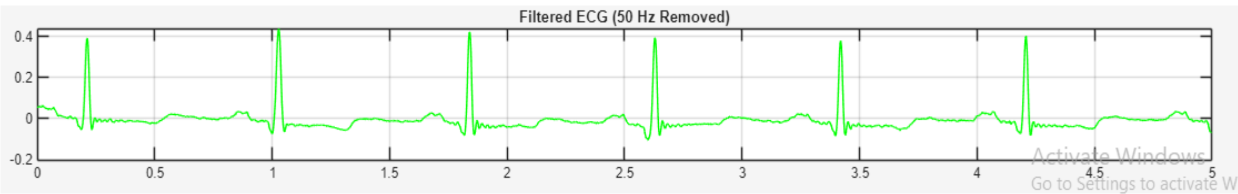


Figure 4.7.1(a): ECG Signal Filtered Using Fir Bandstop Filter Using Hanning Window (Order=200)

Parameter	Value
SNR	15.4286
MSE	0.00018433
RMSE	0.01357672
Overall Gain	0.851111

Table 4.7.1: Performance Parameters of Fir Bandstop Filter Using Hanning Window

4.7.2 Fir Bandstop Filter Using Hamming Window

The FIR filter using Hamming windowing was implemented in MATLAB for Power line Interference noise removal. The filter was designed using the specifications discussed in Section 4.2 and applied to the noisy ECG signal.

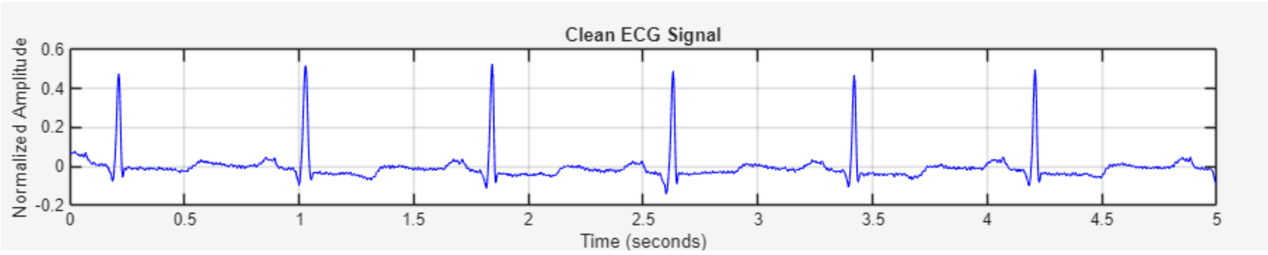


Fig: Original ECG Signal

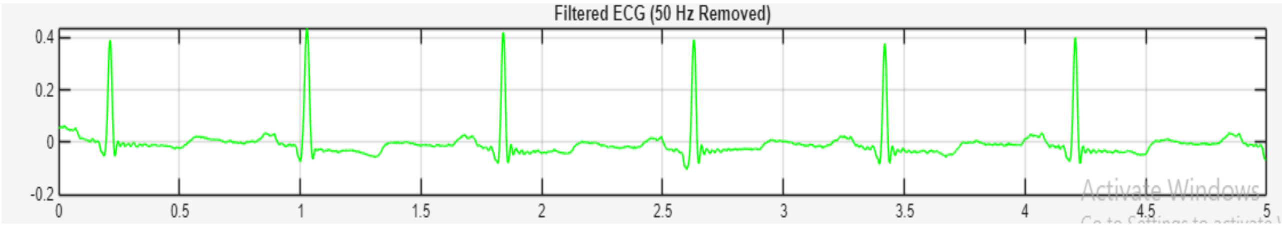


Figure 4.7.2(a): ECG Signal Filtered Using Fir Bandstop Filter Using Hamming Window (ORDER =200)

Parameter	Value
SNR	15.3624
MSE	0.00018716
RMSE	0.01368048
Overall Gain	0.849584

Table 4.7.2: Performance Parameters of Fir Bandstop Filter Using Hamming Window

4.7.3 Fir Bandstop Filter Using Rectangular Window

The FIR filter using Rectangular windowing was implemented in MATLAB for Power line Interference noise removal. The filter was designed using the specifications discussed in Section 4.2 and applied to the noisy ECG signal.

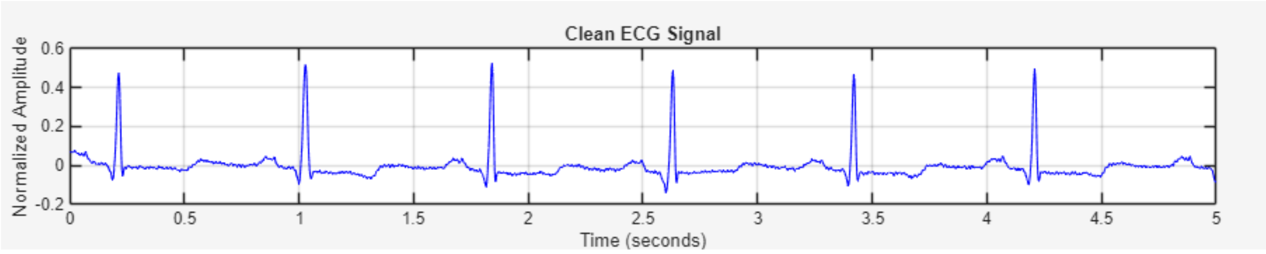


Fig: Original ECG Signal

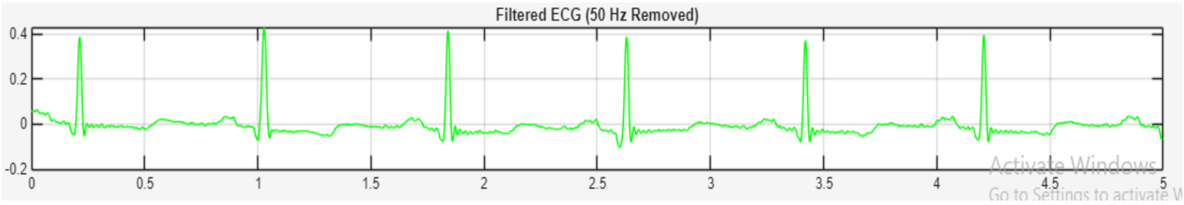


Figure 4.7.3(a): ECG Signal Filtered Using Fir Bandstop Filter Using Rectangular Window (Order =200)

Parameter	Value
SNR	14.6023
MSE	0.00022296
RMSE	0.01493169
Overall Gain	0.832689

Table 4.7.3: Performance Parameters of Fir Bandstop Filter Using

4.7.4 Fir Bandstop Filter Using Blackmann Window

The FIR filter using Blackmann windowing was implemented in MATLAB for Power line Interference noise removal. The filter was designed using the specifications discussed in Section 4.2 and applied to the noisy ECG signal.

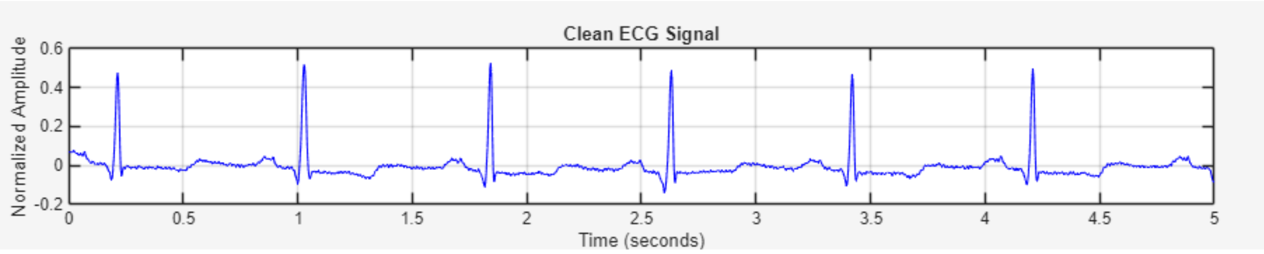


Fig: Original ECG Signal

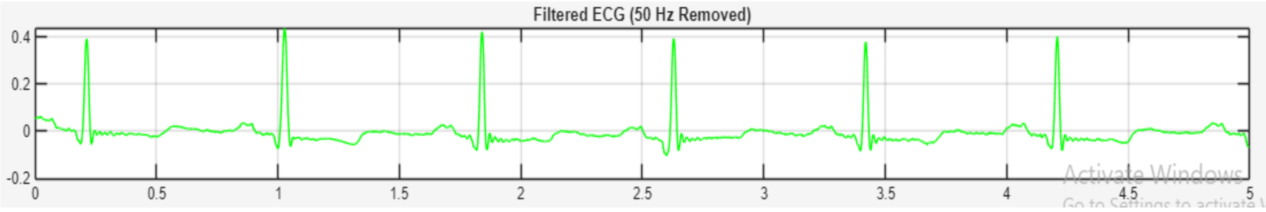


Figure 4.7.4(a): ECG Signal Filtered Using Fir Bandstop Filter Using Blackmann Window (Order =200)

Parameter	Value
SNR	15.4151
MSE	0.00018490
RMSE	0.01359772
Overall Gain	0.850919

Table 4.7.4: Performance Parameters of Fir Bandstop Filter Using Blackmann Window

4.8 DESIGN AND IMPLEMENTATION OF NOTCH FILTERS FOR POWER LINE INTERFERENCE NOISE

A notch filter is actually a special type of band-stop filter. A notch filter rejects only a very narrow frequency band around a specific frequency (such as 50 Hz), whereas a general band-stop filter rejects a much wider range of frequencies.

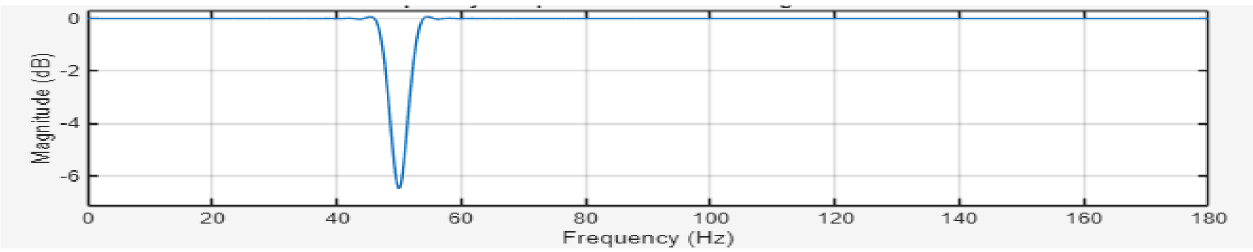


Fig: Frequency response of notch filter

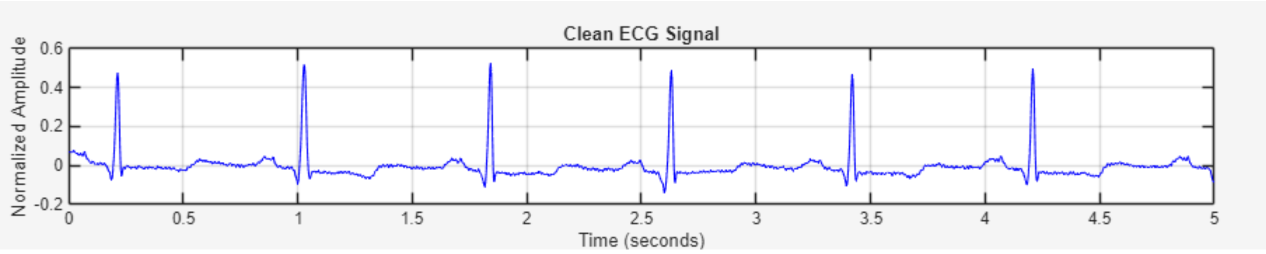


Fig: Original ECG Signal

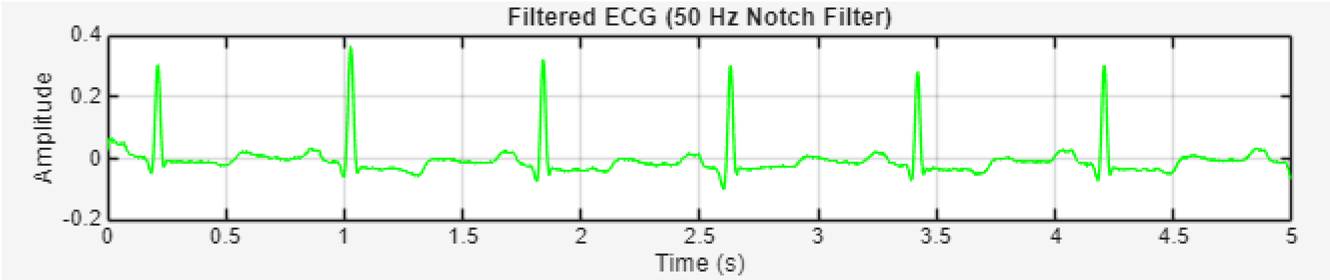


Figure 4.8(a): ECG Signal Filtered Using IIR Notch Filter

Parameter	Value
SNR	10.3324
MSE	0.00059595
RMSE	0.02441210
Overall Gain	0.753239

Table 4.8: Performance Parameters of IIR Notch Filter

OBSERVATIONS FOR PLI NOISE:

1. The Butterworth notch filter achieved the highest SNR since it removed a large amount of interference with minimal signal power loss.
2. Despite its lower SNR, the FIR Hanning filter and IIR Notch filter better preserved the ECG morphology by minimizing phase distortion .
3. For medical ECG applications, waveform preservation is more important than obtaining the highest SNR, as accurate P, QRS, and T wave shapes are essential for diagnosis.
4. The remaining noise after filtering is mainly due to external interference and practical filter implementation (loading) effects.
5. After comparing different filters ,we came to a conclusion that the best filter that can be used to remove PLI noise is FIR filter using hanning windowing method and IIR Notch filter.
6. Comparing the SNR of FIR Hanning filter and IIR Notch filter , we observed that the SNR of hanning filter is high compared to that of IIR notch filter.

7. Also the mean square error of hanning filter is less compared to that of IIR notch filter. From these we can came to conclusion to use FIR Hanning filter for removing PLI noise.
8. Since we are using a bandstop filter we are specifying the bandwidth 0f 20hz with fc1=45Hz and fc2=65Hz.
9. When we take an even narrow bandwidth like 48Hz to 52Hz then the filtering is not much effective compared to that of the previous bandwidth.
10. Also as the order of filter increases the filter reaches ideal filter characteristics, but beyond a certain order the filtering effectiveness reduces.
11. These unnecessary noises are due to external noise and the load effect of the filters.

CHAPTER 5: DSP HARDWARE IMPLEMENTATION

5.1 INTRODUCTION

The implementation of digital filters on hardware is an important step in validating their performance for real-time biomedical signal processing applications. Although MATLAB provides an effective environment for filter design and simulation, hardware implementation is necessary to evaluate the practical feasibility, computational efficiency, and real-time performance of the designed filters.

In this project, the best-performing filters identified for Baseline Wander (BW) and Power Line Interference (PLI) removal are implemented on the TMS320C6713 Digital Signal Processing (DSP) Starter Kit (DSK) using Code Composer Studio (CCS)**.** The filter coefficients and ECG signal data generated in MATLAB are converted into CCS-compatible header files and executed on the DSP processor. The filtered ECG output obtained from the hardware is then compared with the MATLAB simulation results to validate the effectiveness of the selected digital filters.

This chapter presents an overview of the TMS320C6713 DSK, the hardware implementation procedure, the experimental setup, and the real-time execution of the selected filters.

5.2 TMS320C6713 DSK

5.2.1 Overview

The TMS320C6713 Digital Signal Processing Starter Kit (DSK) is a high-performance development platform developed by Texas Instruments (TI) for real-time digital signal processing applications. It is based on the TMS320C6713 floating-point DSP processor, which belongs to the TMS320C6000 family. The board provides a convenient platform for implementing and testing DSP algorithms in real time.

Figure 5.2.1 Photograph of TMS320C6713 DSP Starter Kit

The TMS320C6713 DSK supports 32-bit floating-point arithmetic, making it suitable for applications that require high numerical accuracy, such as biomedical signal processing. In this project, the DSK is used to implement the selected Butterworth and Hanning window filters for ECG signal denoising. The ECG samples and filter coefficients generated in MATLAB are stored as header files and processed on the DSP using Code Composer Studio (CCS)

5.2.2 Hardware Specifications

The important hardware specifications of the TMS320C6713 DSK used in this project are listed in Table 6.1.

Parameter	**Specification**
Processor	Texas Instruments TMS320C6713 DSP
Processor Family	TMS320C6000 Series
Data Type	32-bit Floating Point
Clock Frequency	225 MHz
Architecture	VelociTI™ Very Long Instruction Word (VLIW)
External SDRAM	16 MB
Flash Memory	512 KB
Audio Codec	TLV320AIC23 Stereo Codec

Table – 6.1 Hardware specifications

5.3 ARCHITECTURE OF TMS320C6713 DSK

The architecture of the **TMS320C6713 DSP Starter Kit (DSK)** is shown in **Figure 6.2**. The development board consists of the TMS320C6713 DSP processor, on-board memory, USB JTAG interface, audio codec, CPLD, and peripheral expansion interfaces. These components work together to support the development and execution of digital signal processing applications.

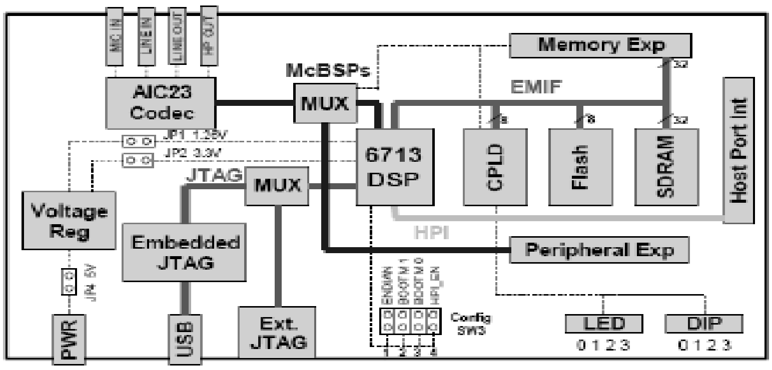


Figure 5.3 Block Diagram of TMS320C6713 DSP Starter Kit (DSK)

The major components of the TMS320C6713 DSK used in this project are described below.

TMS320C6713 DSP PROCESSOR

The TMS320C6713 is the core processing unit of the DSK. It is a 32-bit floating-point DSP processor based on the VelociTI™ Very Long Instruction Word (VLIW) architecture. In this project, the processor executes the selected Butterworth and Hanning window filters to remove Baseline Wander (BW) and Power Line Interference (PLI) from the ECG signal.

SDRAM

The on-board SDRAM provides temporary storage for program execution and data processing. During hardware implementation, the ECG samples, filter coefficients, and filtered output samples are stored in memory while the DSP performs the filtering operations.

FLASH MEMORY

The Flash memory stores boot information and program code required by the DSP. During development, the compiled application is downloaded from Code Composer Studio and executed on the DSP processor.

USB JTAG EMULATOR

The USB JTAG interface establishes communication between the personal computer and the DSP board. It enables downloading of the compiled program, debugging, and execution through Code Composer Studio (CCS).

AUDIO CODEC (TLV320AIC23)

The DSK includes an on-board TLV320AIC23 audio codec for analog-to-digital and digital-to-analog signal conversion. Although the codec supports real-time audio and signal acquisition, it is not utilized in this project because the ECG signal is provided to the DSP as stored digital samples through header files generated in MATLAB.

EXPANSION INTERFACES

The DSK provides Host Port Interface (HPI), External Memory Interface (EMIF), and peripheral expansion connectors for interfacing with external hardware. These interfaces extend the functionality of the board for advanced DSP applications, although they are not used in the present work.

The architecture of the TMS320C6713 DSK provides the necessary computational capability, memory resources, and development support required for implementing digital filtering algorithms. In this project, the DSP processor executes the selected ECG denoising filters using the ECG signal and filter coefficients generated in MATLAB, thereby validating the filter performance on hardware.

5.4 CODE COMPOSER STUDIO (CCS)

5.4.1 Development Environment

Code Composer Studio (CCS) is an Integrated Development Environment (IDE) developed by Texas Instruments for programming, debugging, and executing applications on DSP and embedded processors. It provides a comprehensive platform for developing digital signal processing algorithms by integrating source code editing, project management, compilation, debugging, and hardware communication within a single environment.

In this project, Code Composer Studio Version 7 was used for implementing the selected ECG denoising filters on the TMS320C6713 DSP Starter Kit (DSK). The ECG signal samples and filter coefficients generated in MATLAB were converted into CCS-compatible header files and included in the CCS project. The filtering algorithms were developed in the C programming language and compiled using the CCS compiler. After successful compilation, the executable program was loaded onto the DSP board through the USB JTAG interface for hardware execution.

CCS also provides debugging tools that allow monitoring of program execution, inspection of memory contents, and visualization of output data. These features were utilized to verify the correct implementation of the Butterworth and Hanning window filters and to obtain the filtered ECG output for performance evaluation.

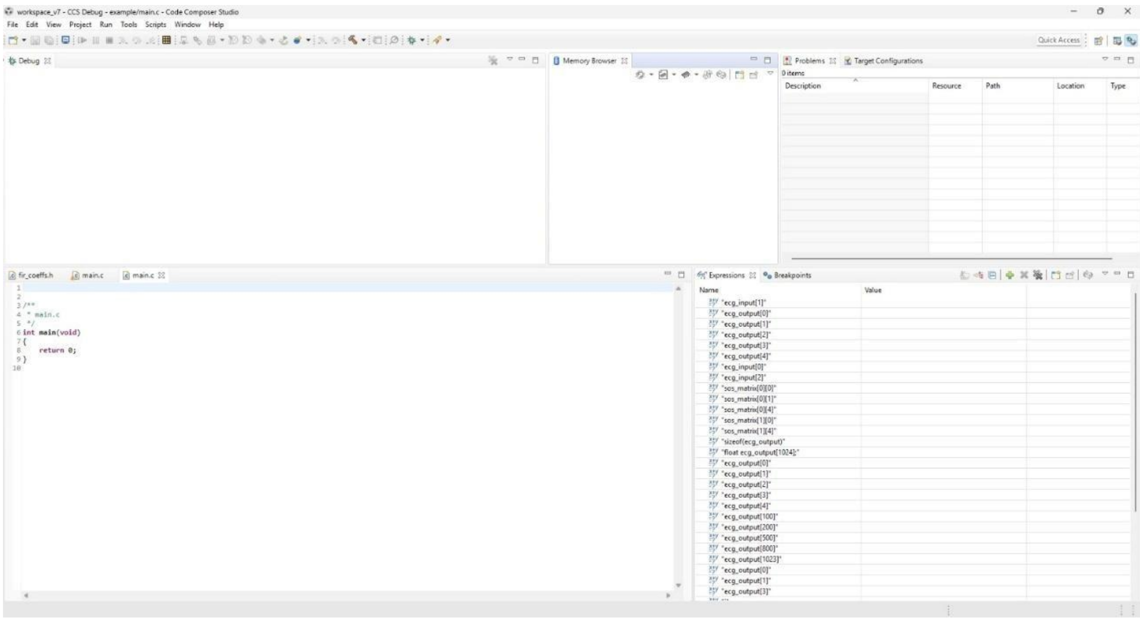


Figure-5.4.1(a) CCStudio Environment

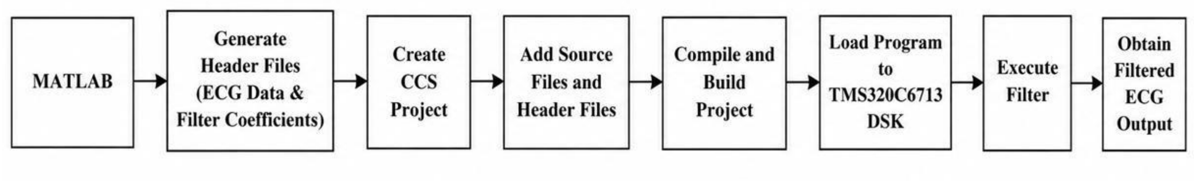


Figure 5.4.1(b): Development Flow Using Code Composer Studio

5.5 HARDWARE IMPLEMENTATION

This section describes the implementation of the selected digital filters on the TMS320C6713 DSK. The ECG signal samples and filter coefficients generated in MATLAB are converted into header files and incorporated into the Code Composer Studio (CCS) project. The Butterworth IIR filter and Hanning window FIR filter are then executed on the DSP processor to obtain the filtered ECG output.

5.5.1 Inclusion of Header Files in CCS

The ECG signal samples and the filter coefficients generated in MATLAB were exported as C header files. These header files were added to the Code Composer Studio (CCS) project and served as the input data for hardware implementation. The ECG samples were stored in `ecg_input.h`, while the filter coefficients were stored in `iir_coeffs.h` and `fir_coeffs.h` for the Butterworth and Hanning window filters, respectively.

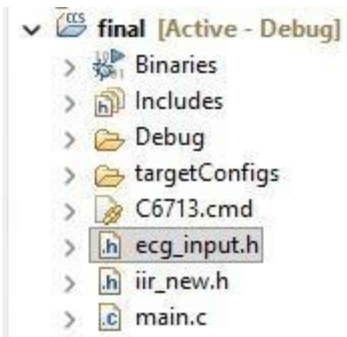


Fig 5.5.1: CCS Project with Source and Header Files

5.5.2 Butterworth Filter Implementation for Baseline Wander Noise removal

The Butterworth high-pass filter selected during MATLAB analysis was implemented on the **TMS320C6713 DSP Starter Kit (DSK)** for Baseline Wander removal.

The ECG input data (`ecg_data.h`) and the Butterworth filter coefficients (`iir_coeffs.h`) generated in MATLAB were imported into the **Code Composer Studio (CCS)** project along with the C source files (`main.c` and `butter.c`).

During execution, the `main.c` program sequentially reads the ECG samples and calls the **ButterworthHPF()** function, while `butter.c` performs the filtering operation using the Butterworth.

filter coefficients stored in the SOS matrix.

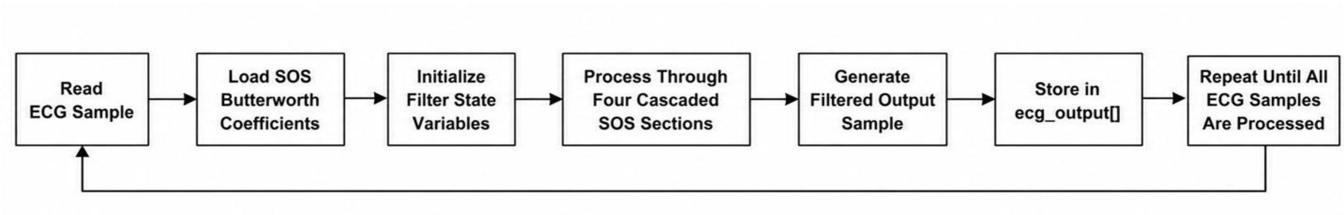


Figure 5.5.2: Butterworth Filter Processing Flow on the TMS320C6713 DSP

After successful compilation, the **Spectrum Digital DSK-EVM-eZdsp Onboard USB Emulator** is selected to establish communication between the computer and the **TMS320C6713 DSP Starter Kit (DSK)**.

The executable program is then downloaded to the DSP board through the emulator. During program loading, the onboard LEDs blink briefly, indicating successful program transfer and initialization. Breakpoints are inserted at appropriate locations to verify program execution and inspect the input and output data during debugging.

After successful verification, the program is executed and the DSP processes all ECG samples using the Butterworth filter coefficients to generate the filtered ECG output. The output waveform is displayed using the CCS Graph utility and compared with the MATLAB simulation results to validate the hardware implementation.

5.5.3 Hanning Window Filter Implementation for PLI noise removal

The Hanning window-based FIR filter selected during MATLAB analysis was implemented on the **TMS320C6713 DSP Starter Kit (DSK)** for the removal of Power Line Interference (PLI).

The ECG input data (`ecg_data.h`) and the FIR filter coefficients (`fir_coeffs.h`) generated in MATLAB were imported into the **Code Composer Studio (CCS)** project along with the C source files (`main.c` and `fir.c`). During execution, the `main.c` program sequentially reads the ECG samples and calls the **FIR_Filter()** function, while `fir.c` performs the filtering operation using the Hanning window FIR filter coefficients stored in the coefficient array.

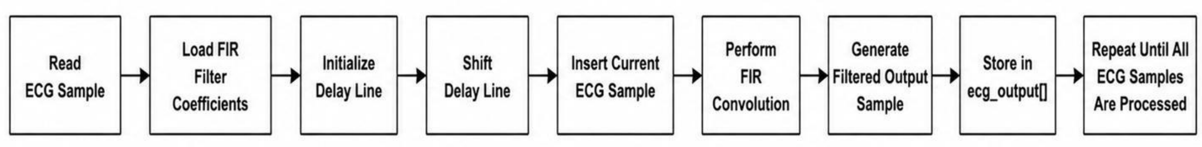


Figure 5.5.3: Hanning Window FIR Filter Processing Flow on the TMS320C6713 DSP

After successful compilation, the **Spectrum Digital DSK-EVM-eZdsp Onboard USB Emulator** is selected to establish communication between the computer and the **TMS320C6713 DSP Starter Kit (DSK)**.

The executable program is then downloaded to the DSP board through the emulator. During program loading, the onboard LEDs blink briefly, indicating successful program transfer and initialization. Breakpoints are inserted at appropriate locations to verify program execution and inspect the input and output data during debugging.

After successful verification, the program is executed and the DSP processes all ECG samples using the Hanning window FIR filter coefficients to generate the filtered ECG output. The output waveform is displayed using the CCS Graph utility and compared with the MATLAB simulation results to validate the hardware implementation.

5.6 HARDWARE OUTPUT EVALUATION

This section presents the output waveforms obtained from the TMS320C6713 DSP Starter Kit after implementing the Butterworth and Hanning window filters. The filtered ECG signals are displayed using the Code Composer Studio (CCS) Graph utility to verify the successful execution of the filters on the DSP hardware.

5.6.1 Butterworth Filter Output for BW noise

Figure 6.8 shows the ECG waveform obtained after implementing the Butterworth high-pass filter on the TMS320C6713 DSP Starter Kit (DSK). The hardware output is presented along with the corresponding clean ECG signal to evaluate the performance of the implemented filter for Baseline Wander removal. The following observations were made from the obtained hardware output.

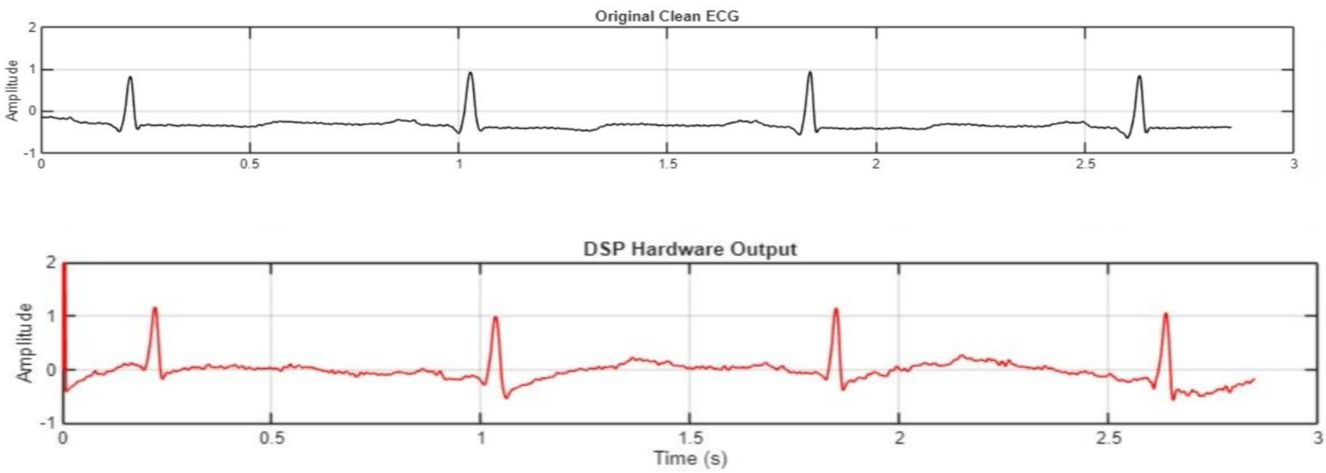


Figure 5.6.1: Ecg waveform and Dsp Hardware output

OBSERVATIONS:

- The Butterworth filter was successfully implemented and executed on the TMS320C6713 DSP processor.
- The QRS complexes are clearly identifiable in the hardware output, indicating that the major ECG peaks are retained after filtering.
- The hardware output exhibits residual baseline fluctuations, indicating that complete removal of Baseline Wander was not achieved.

- Variations in the baseline level and waveform amplitude are observed when compared with the clean ECG signal.
- The hardware implementation produced an **MSE of 0.2313** and an **RMSE of 0.4809**, indicating a measurable deviation between the hardware output and the reference ECG signal.
- Despite these deviations, the hardware implementation successfully demonstrated the execution of the Butterworth filter on the DSP platform and generated a valid filtered ECG waveform.

5.6.2 Hanning Window Filter Output For Pli Noise Removal

Figure 6.9 shows the ECG waveform obtained after implementing the Hanning window FIR filter on the TMS320C6713 DSP processor. The raw hardware output is compared with the reference ECG signal, and the processed waveform obtained after scaling and shifting is also presented for better visualization of the ECG characteristics.

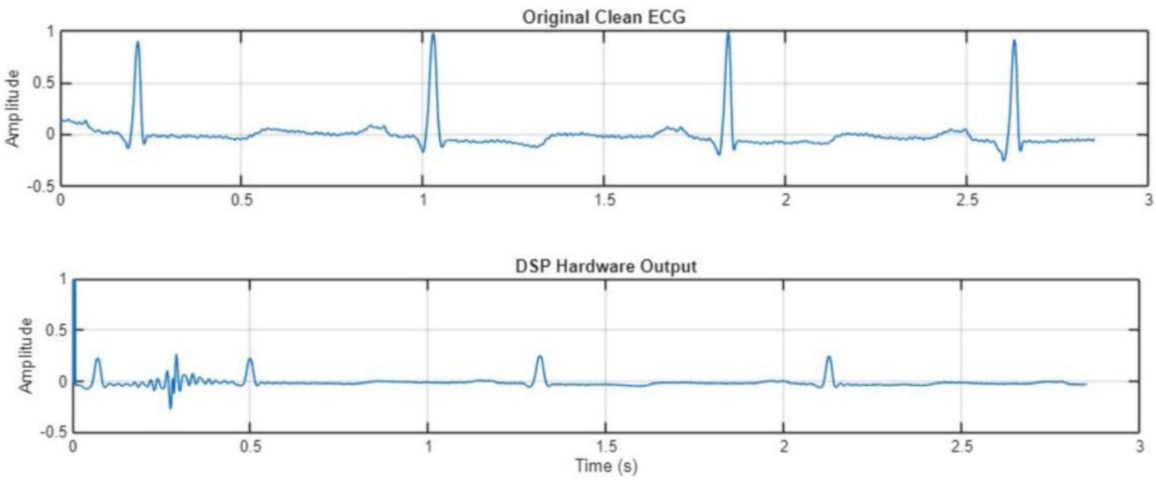


Figure 5.6.2(a): - Raw hardware output

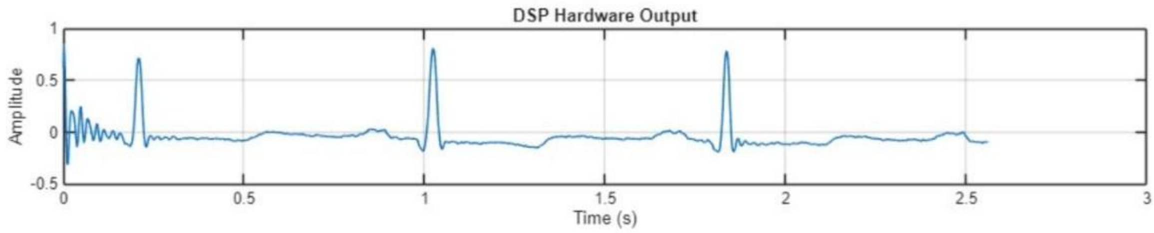


Figure 5.6.2(b): - Scaled and Shifted Hardware Output

OBSERVATIONS

- The Hanning window FIR filter was successfully implemented and executed on the DSP processor.
- The raw hardware output exhibits a lower signal amplitude than the reference ECG waveform.
- A short transient/noise is observed at the beginning of the hardware output before the signal reaches a stable response.

DELAY CALCULATION IN FIR FILTERS:

Since the total number of filter coefficients in our FIR filter is 201 , the delay becomes,

$$\boxed{\text{DELAY} = (N-1)/2}$$

DELAY = (201-1)/2 =100 samples

The sampling frequency in our clean ECG signals , Fs = 360.

Therefore the time, T = (1/Fs) = (1/360) sec.

So we can say that for every (1/360) seconds, we get 1 sample.

For 200 samples, we get 0.277 seconds

So by shifting the hardware ECG output signal by 0.277 sec backwards, we can actually see that the original signal is matched with the original ECG signal.

We can also conclude that the filtering is done successfully.

- After applying scaling and shifting for visualization, the major ECG features, including the QRS complexes, become more clearly observable.
- The waveform obtained after scaling follows the overall shape of the reference ECG more closely, although minor amplitude differences remain.
- The hardware implementation produced an **MSE of 0.2313** and an **RMSE of 0.4809**, indicating a measurable deviation between the hardware output and the reference ECG signal.

The hardware output obtained from the DSP processor exhibits lower amplitude than the reference ECG signal. The scaled waveform demonstrates that the ECG information is preserved, while the reduced output amplitude is primarily associated with the hardware implementation rather than the filtering algorithm. For practical ECG acquisition systems, an additional analog output amplification stage may be employed after the DSP processor to restore the signal to the desired amplitude level. The transient observed at the beginning of the output is attributed to the filter initialization before the delay elements reach steady-state operation.

CHAPTER-6 RESULTS AND DISCUSSION

6.1 INTRODUCTION

This chapter presents the results obtained from the MATLAB simulation and the TMS320C6713 DSP hardware implementation of the proposed ECG denoising system. The performance of the selected Butterworth high-pass filter and Hanning window FIR filter is evaluated by comparing the software and hardware outputs. The comparison is based on ECG waveforms and performance parameters such as Mean Square Error (MSE) and Root Mean Square Error (RMSE), followed by a discussion of the observed results.

6.2 BASELINE WANDER REMOVAL RESULTS

Figure 6.1 presents the comparison of the clean ECG signal, the MATLAB-filtered ECG, and the DSP hardware output obtained using the Butterworth high-pass filter for Baseline Wander removal. The MATLAB implementation provides the reference filtered output, while the DSP output demonstrates the practical performance of the filter on the TMS320C6713 DSP processor.

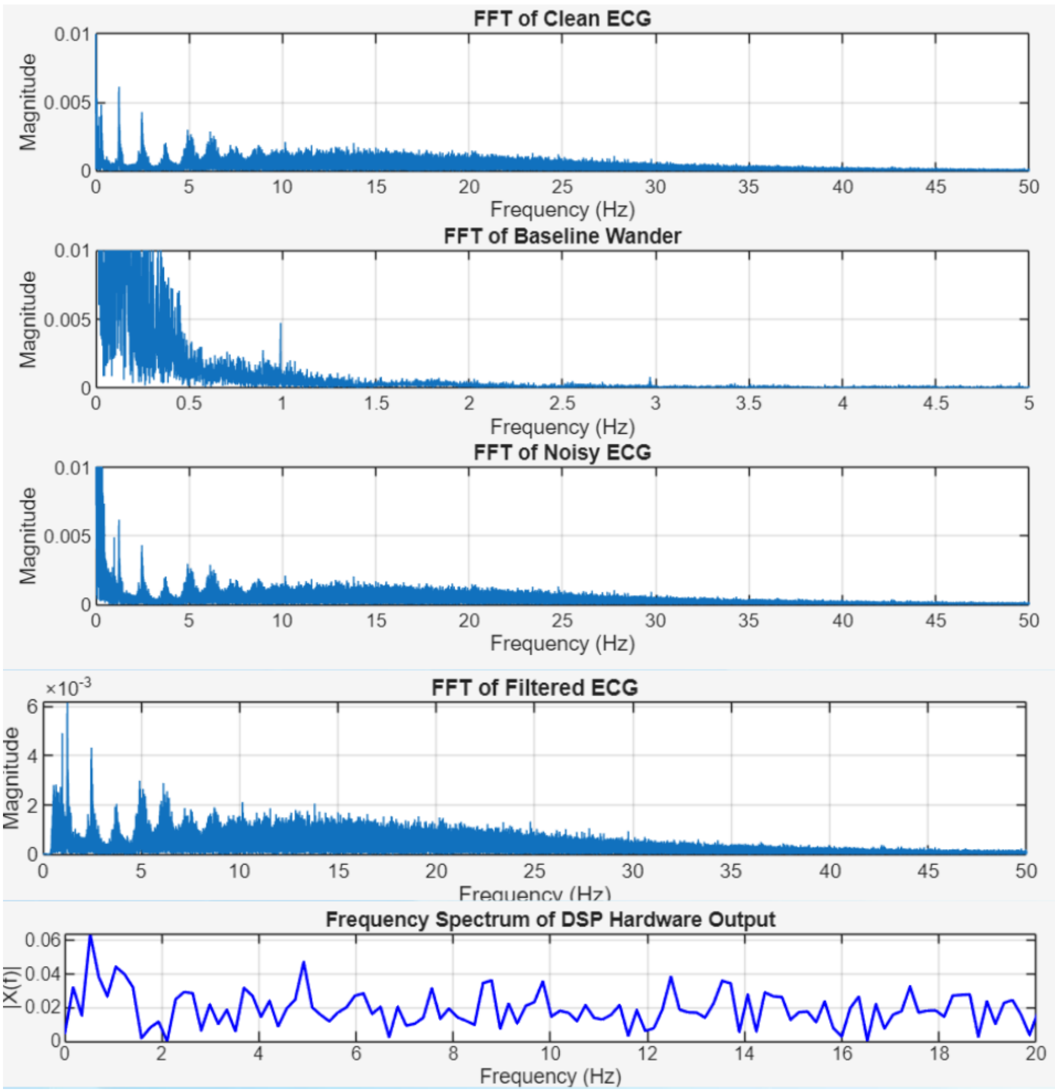


Figure 6.2.2: Comparison of Clean ECG, MATLAB Output frequency spectrums

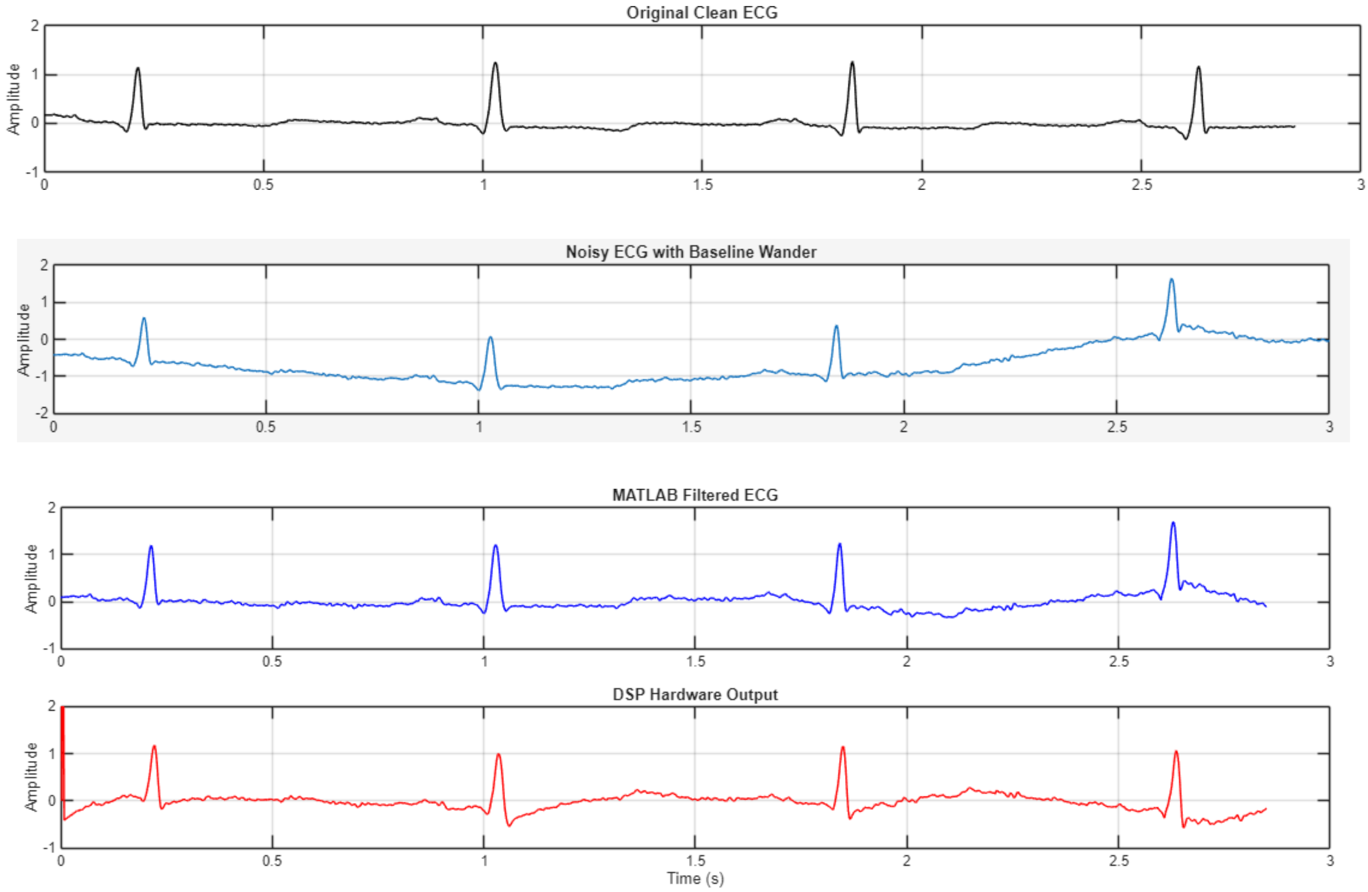


Figure 6.2.1: Comparison of Clean ECG, ECG+NOISE, MATLAB Output, and DSP Hardware Output for Baseline Wander Removal

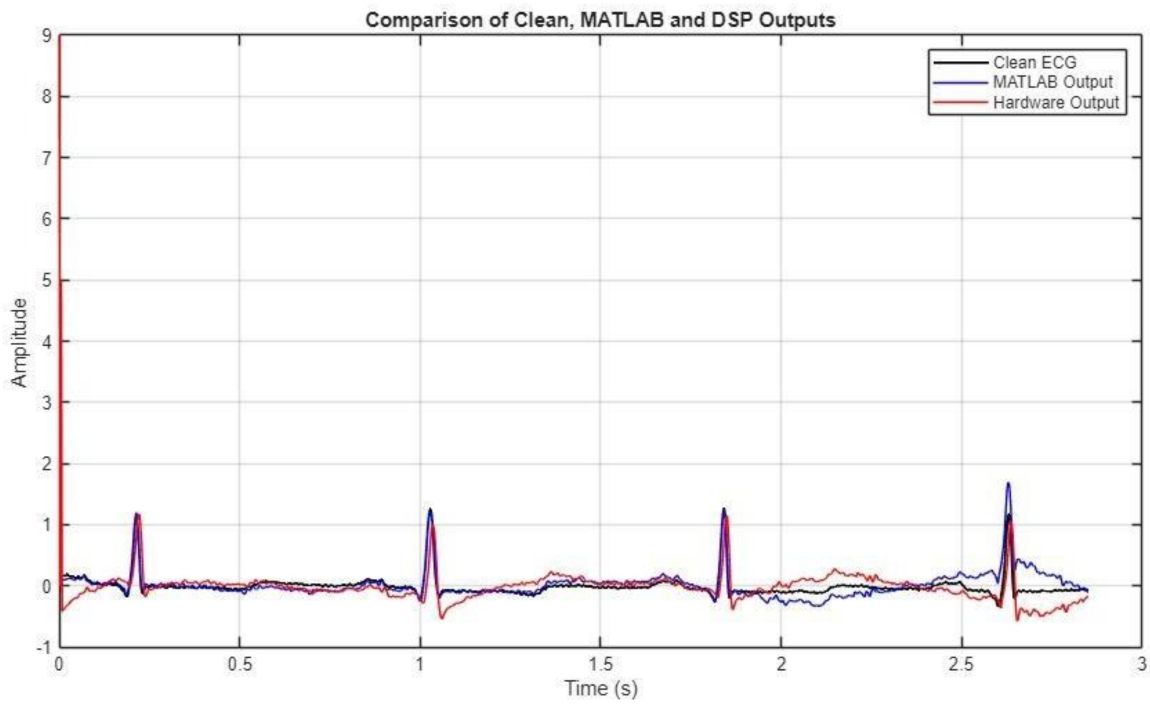


Figure- 6.2.2 Combined comparison plot (Clean ECG, MATLAB Output, Hardware Output)

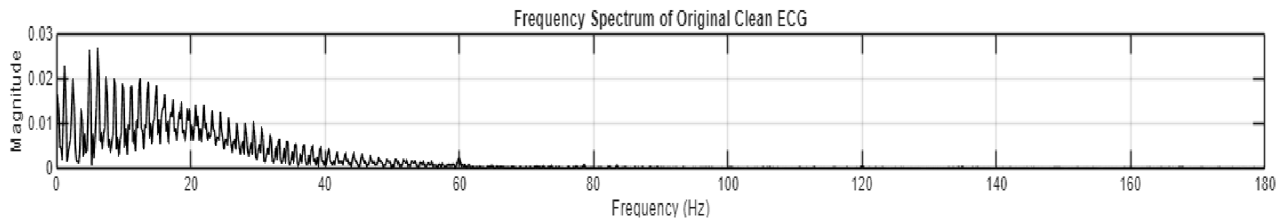
Observations:

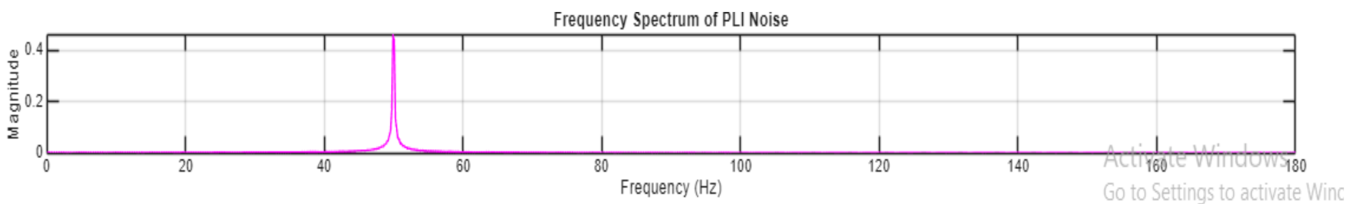
- Both the MATLAB and DSP implementations successfully execute the Butterworth high-pass filter for Baseline Wander removal.
- The MATLAB output exhibits better baseline correction and produces a waveform that more closely resembles the clean ECG signal.
- The DSP hardware output preserves the major QRS complexes; however, residual baseline fluctuations are observed.
- The hardware output shows slight amplitude variations compared with the MATLAB output.
- The MATLAB implementation achieves lower error values than the DSP hardware implementation, indicating closer agreement with the clean ECG signal.
- The differences between the software and hardware outputs can be attributed to practical hardware implementation effects, resulting in a slight increase in reconstruction error.
- SNR of clean ECG = -10.7674

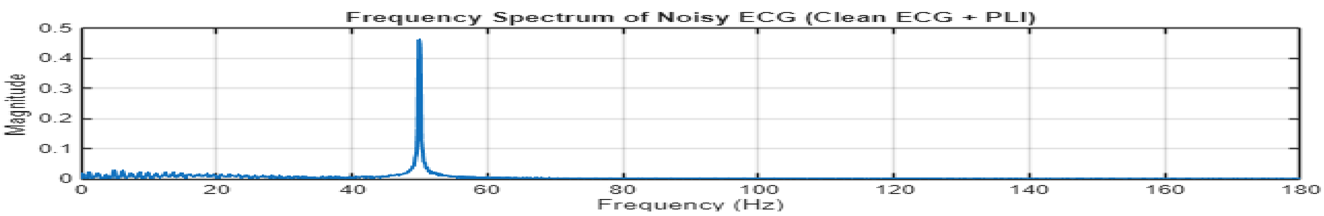
Performance Metric	MATLAB results	DSP Hardware results
SNR	1.1084	2.4843
MSE	0.1315	0.2313
RMSE	0.3627	0.4809

6.3 POWER LINE INTERFERENCE REMOVAL RESULTS

Figure 6.3.2 presents the comparison of the clean ECG signal, the MATLAB-filtered ECG, and the processed DSP hardware output obtained using the Hanning window FIR filter for Power Line Interference (PLI) removal. The hardware output shown has been appropriately scaled and shifted to facilitate comparison with the software results. Figure 6.5 provides a combined comparison of the clean ECG, MATLAB output, and DSP hardware output.







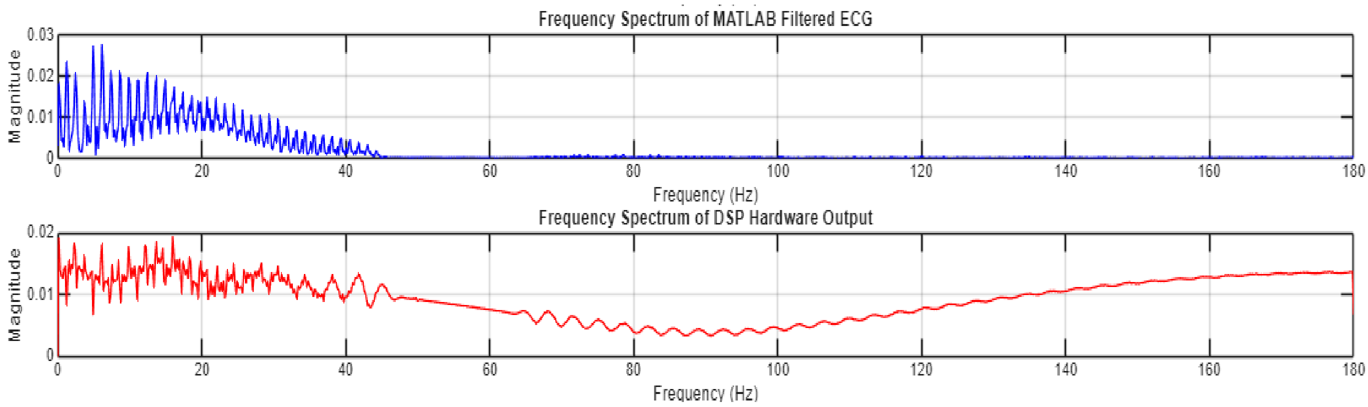


Fig: frequency response of Clean ECG, Noise, Clean+Noise, Matlab filtered, Hardware filtered

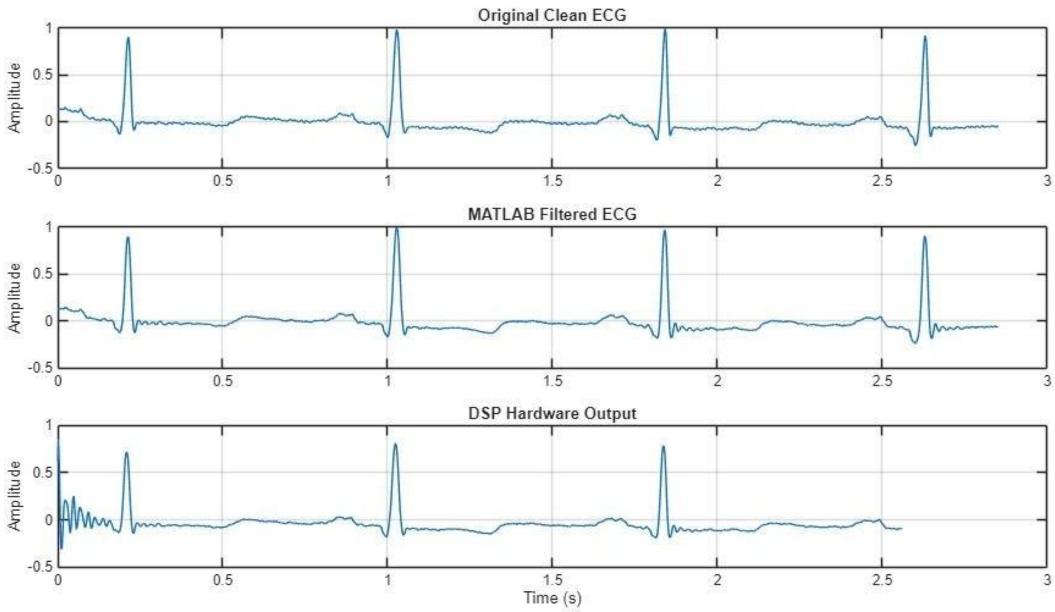


Figure 6.3.1 – Original ECG, MATLAB Filtered ECG, and Shifted & Scaled DSP Hardware

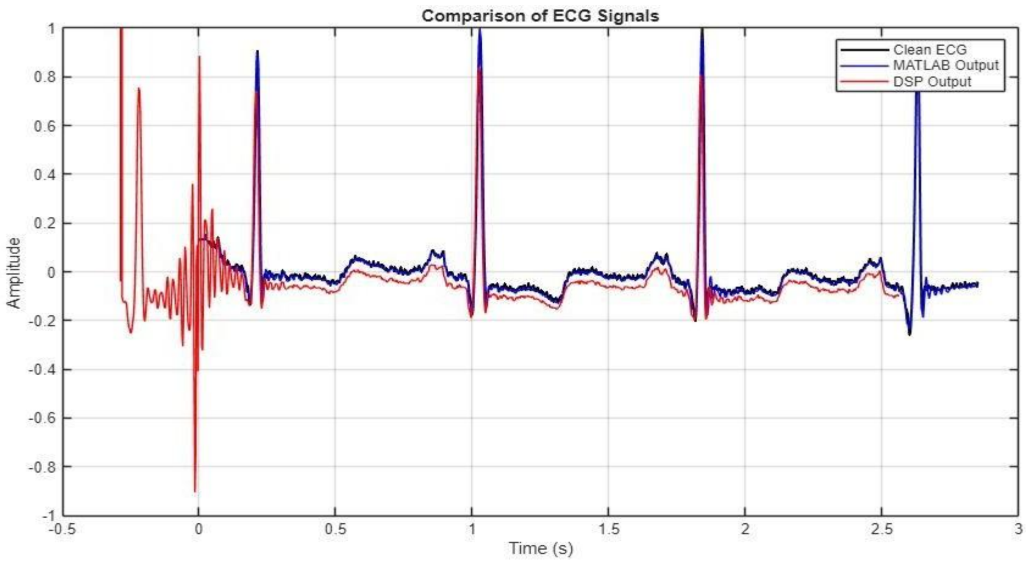


Figure 6.3.2 – Combined comparison plot (Clean ECG, MATLAB Output, Hardware Output)

Observations

- Both the MATLAB and DSP implementations successfully suppressed the 50 Hz Power Line Interference.
- The MATLAB-filtered ECG closely resembles the clean ECG waveform, indicating effective removal of the interference.
- The processed DSP hardware output follows the overall trend of the MATLAB-filtered ECG, with the major QRS complexes clearly preserved.
- Minor amplitude variations are observed in the hardware output when compared with the MATLAB result.
- The hardware implementation produces slightly higher MSE and RMSE values than the MATLAB implementation, indicating a small deviation from the software output.
- Overall, the Hanning window FIR filter demonstrates satisfactory performance on the DSP hardware for real-time Power Line Interference removal.
- SNR of Clean ECG = -10.7674

Performance Metric	MATLAB results	DSP Hardware results
SNR	15.4286	-7.8074
MSE	0.000213	0.1221
RMSE	0.014796	0.1349

6.4 OVERALL DISCUSSION

The comparative analysis demonstrates that both the Butterworth high-pass filter and the The results obtained from both the MATLAB simulation and the DSP hardware implementation demonstrate the effectiveness of the proposed ECG denoising system for the removal of Baseline Wander (BW) and Power Line Interference (PLI). The Butterworth high-pass filter and the Hanning window FIR filter, identified as the best-performing filters during the software evaluation, were successfully implemented on the TMS320C6713 DSP Starter Kit (DSK).

For Baseline Wander removal, the MATLAB implementation provided better baseline correction and lower error values than the hardware implementation. Although the DSP output preserved the major ECG features, slight baseline fluctuations and amplitude variations were observed, resulting in comparatively higher MSE and RMSE values.For Power Line Interference removal, both the MATLAB and DSP implementations effectively suppressed the 50 Hz interference while preserving the major ECG waveform characteristics. The processed hardware output closely followed the MATLAB-filtered ECG, with only minor differences in signal amplitude. The hardware implementation exhibited slightly higher error values than the software implementation due to practical hardware implementation factors.

Overall, the MATLAB implementation achieved superior denoising performance under ideal simulation conditions, while the DSP hardware implementation successfully demonstrated real-time execution of the selected filters. The obtained results confirm that the proposed ECG denoising system is suitable for real-time biomedical signal processing applications, with only minor deviations between the software and hardware implementations.

CHAPTER 7: CONCLUSION AND FUTURE SCOPE

7.1 CONCLUSION

The present work successfully demonstrated the design, simulation, and real-time implementation of digital filters for ECG signal denoising using the TMS320C6713 DSP processor. Two major ECG noise components, namely baseline wander and power line interference, were considered for evaluation. Different FIR and IIR filter structures were designed and analyzed in MATLAB using objective performance parameters such as Signal-to-Noise Ratio (SNR), Mean Square Error (MSE), and gain

Based on the simulation results, the 8th-order Butterworth high-pass IIR filter was identified as the most suitable filter for baseline wander removal, while the 201-tap Hamming window-based FIR bandstop filter provided the best performance for suppressing 50 Hz power line interference. The selected filters effectively removed the corresponding noise components while preserving the essential morphological features of the ECG signal, including the P, QRS, and T waves.

The optimized filter coefficients were implemented on the TMS320C6713 DSP processor using Code Composer Studio, and the hardware outputs were compared with the MATLAB simulation results. Although minor variations in amplitude and transient response were observed due to practical hardware limitations such as finite-precision arithmetic, initialization effects, and real-time processing constraints, the overall filtering performance closely matched the software implementation. This confirms the correctness of the filter design and demonstrates the feasibility of implementing real-time ECG denoising algorithms on DSP hardware.

Overall, the project achieved its objective of developing an efficient ECG denoising system capable of improving signal quality through both software simulation and hardware implementation, making it suitable for real-time biomedical signal processing applications.

7.2 FUTURE SCOPE

The present work can be extended in several directions to further improve ECG denoising performance and support advanced biomedical applications. Possible future enhancements include:

- Implementation of adaptive filtering techniques such as LMS and RLS for dynamic noise environments.
- Investigation of wavelet transform and hybrid filtering methods for improved suppression of multiple ECG noise sources.
- Optimization of the filtering algorithms to reduce computational complexity and improve real-time processing efficiency.

- Implementation on modern embedded platforms such as FPGA, ARM-based processors, or SoC devices for portable healthcare applications.
- Integration of wireless communication modules for remote ECG monitoring and telemedicine systems.
- Addition of an output signal conditioning or amplification stage to improve the amplitude of the hardware output for practical biomedical instrumentation.

REFERENCES

[1] LEIF SO ¨RNMO Lund University Sweden PABLO LAGUNA Zaragoza University Spain-ELECTROCARDIOGRAM (ECG) SIGNAL ECG PROCESSING.

[2] Accurate ECG Signal Processing By Ajay Bharadwaj, Applications Engineer Sr, and Umanath Kamath, Contingent Workforce, Cypress Semiconductor Corp.

[3] USING OF FIR AND IIR FILTERS FOR NOISE REMOVAL FROM ECG SIGNAL: A PERFORMANCE ANALYSIS Gandham Sreedevi Dept. of ECE, Sri Venkateswara University College of Engineering, Tirupati-517502, Andhra Pradesh, India

[4] Denoising of ECG Signals Using FIR & IIR Filter: A Performance Analysis Ms. Chhavi Saxena1 , Mr. Vivek Upadhyaya2 , Dr. Hemant Kumar Gupta3 Dr. Avinash Sharma4

[5] Baseline Wandering Removal in ECG Signal Using Filters 1Dr. M. Shanmukhi, 2Kuruva Harinath, Nazia Tabassum, 4Mohammed Mubeen